

Online Appendix

A1 Joining the PKV System and Enrolling in GLTHI

As mentioned in Section 2, the decision to join the PKV is essentially a lifetime decision. The basic social insurance principle is: “Once private, always private[ly insured]” (Schencking, 1999; Jacobs et al., 2012; Innungskrankenkasse Berlin Brandenburg, 2018). It is only possible to switch back to GKV under very restrictive circumstances, see Section A1.1 below. The institutional reason is to avoid that the young and healthy switch to PVK and pay (lower) income-*independent* premiums, and switch back to GKV when they are old and sick later in life to pay (lower) income-*dependent* GKV contribution rates.^{A52}

People leave the GKV permanently to join the PKV for several reasons: First, PKV premiums are actuarially fair instead of income-dependent contribution rates. On the other hand, GKV offers free family coverage for non-working relatives. This implies that childless and healthy high-income earners tend to be better off financially in the PKV.

Second, in the PKV, people choose between thousands of plans which they customize according to individual preferences and needs. Applicants can choose their benefit level and cost-sharing amounts, within some lax regulatory limits.^{A53} Most insurers operate nationwide, are open to all applicants, and sell policies in all states. Policies are also portable across state lines.

Third, the PKV relies on private contracts with one-sided commitment over the lifecycle. By contrast, GKV pricing and the essential benefit package depend inherently on the political and economic situation and funding constraints. One could argue that the public system carries more uncertainty about future benefits and contribution rates. Below, we discuss the GLTHI lifecycle premium calculation in theory and practice.

Fourth, regulation makes the PKV particularly attractive for civil servants. The large majority enrolls in GLTHI. Since the 1920s, German law stipulates that the state has the obligation to care for their civil servants (*Fürsorgepflicht*), which is why the government self-insures (at least) 50 percent of their civil servants’ health care costs directly (*Beihilfe*).^{A54} However, in general, the government

^{A52}Obviously, the advantage of having income-independent premiums can backfire when income drops later in life. For financial hardship cases, the regulator introduced the Basic Plan, see footnote A61.

^{A53}For example, effective 2009, the government mandated that deductible cannot exceed €5,000 for new policies (see *GKV-Wettbewerbsstärkungsgesetz*).

^{A54}In practice, this means that civil servants first have to pay providers out-of-pocket and then submit their claims to a state agency *and* the private insurer, both of which then reimburse half the costs. This principle that the insured must first pay providers and then submit claims (*Kostenerstattungsprinzip*) is a general GLTHI principle and downside for policyholders. Compared to GKV, where providers directly submit claims to sickness funds and cost-sharing is low, it

does not pay half the GKV contribution rates as private employers do (Schmidt, 2024). This implies that civil servants would have to pay the full GKV contribution rate, but only insure 50 percent of their health care costs on the PKV market.^{A55}

Finally, while both GKV and PKV cover all “medically necessary” benefits and the medical treatment quality is similar, waiting times in the outpatient sector are shorter for the privately insured as outpatient reimbursement rates are structurally higher (Werbeck et al., 2021). Thus, the privately insured are more profitable for providers, also because some PKV benefit packages are more generous than the GKV essential benefit package. As a consequence, PKV insured enjoy higher service quality (but potentially also overtreatment) by physicians.

The fact that some population subgroups can choose between GKV and PKV has not only been criticized on the ground of fairness, but it implies selection into the PKV system. Specifically, as initial premiums are risk-rated, those who have pre-existing conditions and are unhealthy likely decide to stay in the GKV. As discussed, income, family status, and spouse’s labor supply decisions also determine the decision to sign a GLTHI contract. Further, preferences and risk tolerance may determine the decision to opt out of GKV.

A1.1 Switching from PKV to GKV

There exist very limited institutional exemptions for PKV insured to switch to the public GKV system (cf. Netzwerk der Verbraucherzentralen in Deutschland, 2024). Below we provide empirical evidence on the switching rates.

First, for PKV insured from age 55, switching to GKV is essentially impossible, even when their incomes decrease substantially, or when they become unemployed. From age 55, one option is to exit the labor force and enroll under a GKV family plan of a spouse, if available. Rules for joining the GKV have been very strict for older employees. As mentioned, this is to avoid that individuals join the private system when young and healthy with low risk-rated premiums, and switch back to the public system when old and sick with low income (and thus low income-dependent contribution rates).

Second, PKV insured employees below age 55 can only join the public GKV system if they become unemployed or their gross wage permanently decreases below the income threshold (Ver-

implies more paperwork, uncertainty about claim denials, and requires more liquidity.

^{A55} The exact share of *Beihilfe* depends on the state, see Schmidt (2024) for details. Several private insurers have specialized in providing customized coverage for the remaining uninsured share. One out of 16 states, Bremen also covers half of civil servants’ GKV contribution rate which implies giving them the choice between GKV and PKV. The other states only approve such requests in exceptional cases.

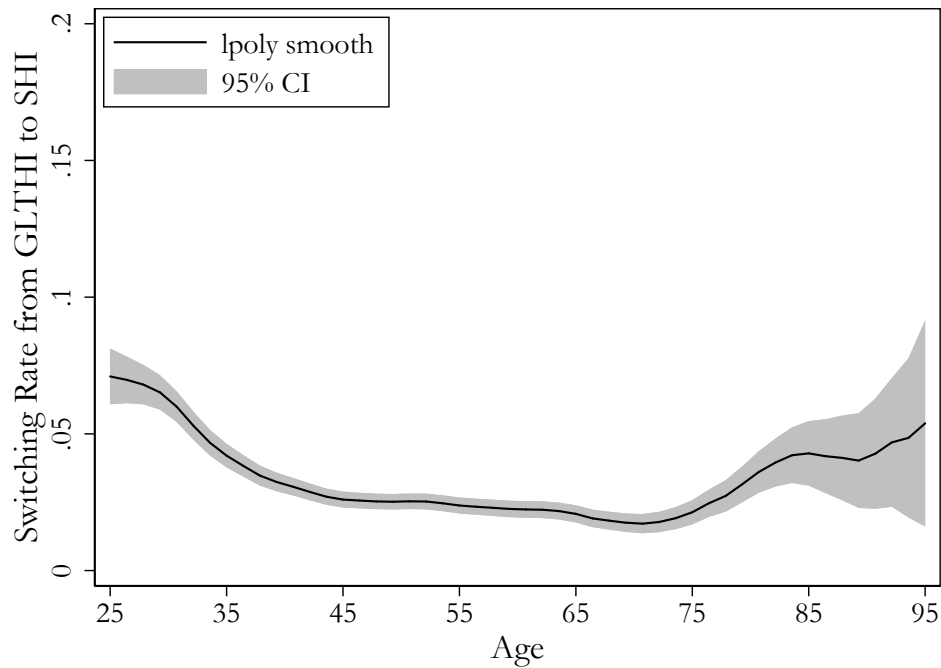


Figure A1: Likelihood to Return to GKV by Age

Notes: The figure shows the probability of returning from the private (PKV) to the public (GKV) system, by age, smoothed with an Epanechnikov kernel, degree 0, bandwidth 2.6. Source: Own calculations using [SOEP \(2018\)](#).

sicherungspflichtgrenze).^{A56} Moreover, permanently switching to the GKV implies losing the entire old-age provisions with averages of €36,108 per policyholder in 2022 ([Association of German Private Healthcare Insurers, 2023](#)).

Third, PKV insured who are self-employed and below the age of 55 can only switch to GKV if they give up their business and become an employee with a gross salary below the income threshold (see Social Code Book V, Para. 6 for details of the law, [Büser, 2012](#); [Cecu, 2018](#)).

Data from the Association of German Private Healthcare Insurers (*Verband der Privaten Krankenversicherung*) show that 124,900 individuals, or 1.4 percent of all PKV insured, switched from PKV to GKV and 145,000 switched from GKV to PKV ([Association of German Private Healthcare Insurers, 2024a](#)). Since 1997, the number of switchers to the GKV system has been very stable between 124,900 and 154,800 per year ([Bundesministerium für Gesundheit, 2024](#)).^{A57} Figure A1 uses representative

^{A56} Assuming an average annual premium of €3,900 (as observed in our data), for an equally high GKV premium (15.5% of the gross wage), annual labor income would need to be as low as €25,000. Hence, artificially reducing income just to be able to join the GKV does not make sense for the overwhelming majority of cases. Further, as stated in Section 2, short-term income manipulation to join the GKV is not allowed ([Krankenkasse, 2024](#)).

^{A57} The total number of PKV insured increased from in 1991 to 8.976.400 in 2011. Since then, it has decreased slightly to 8.704.500 in 2023 ([Bundesministerium für Gesundheit, 2024](#); [Association of German Private Healthcare Insurers, 2024a](#)). Several reforms in the last decades have at least partly determined switching rates over time: The *Gesundheitsreformgesetz* of December 20, 1988 substantially tightened the possibility of joining the PKV for pensioners; the *Gesundheitsstrukturgesetz*, passed on December 21, 1992, introduced the free choice of GKV sickness funds, along with other provisions about the regulation of private insurers. Due to these and other reforms (e.g. the *GKV-Wettbewerbsstärkungsgesetz* of 2007), the GKV-PKV switching rate as a share of all GLTHI policyholders has declined over time.



Figure A2: Age Distribution of Initial Plan Inception

Notes: The figure shows the age distribution for individuals who first join the private insurance company. Source: German Claims Panel Data.

SOEP data to plot switching rates by age. As seen, the likelihood to return to GKV decreases substantially between the age of 25 and 35. We conjecture that this is mostly because those who were privately insured as students enter the labor market and have to enroll in GKV if their gross wages are below the mandatory income threshold. Switching rates remain stable at a low level between age 40 and age 75, and then slightly increase again. Using a fixed effects regression for the probability of switching to GKV among the universe of Germans who were at least once GLTHI policyholder, we find few significant predictors of switching back to GKV. In particular, health care utilization (number of hospital nights and doctor visits) are not significant predictors. The results of this analysis are available upon request.

A2 GLTHI Premium Calculation: Further Details

This section first summarizes the institutional details of the PKV lifecycle premium calculation. Hofmann and Browne (2013) and Atal et al. (2019) provide English references on the topic.

A2.1 Graphical Depiction of Basic Principles

Figure A3 illustrates the principle of constant lifecycle premiums (Rosenbrock, 2010; Hofmann and Browne, 2013). It uses our claims data as input and showcases four risk-age combinations at initial enrollment: low vs. high health risks, and initial enrollment at 30 vs. 50 years. This illustration assumes constant health risk types over the lifecycle.^{A58} The low risk type (the “healthy”) corresponds to a hypothetical individual with no pre-existing conditions; we denote her expected lifecycle health expenditures conditional on survival as $E(m|\text{surv}, \text{low})$. The high risk type (the “sick”) corresponds to a hypothetical individual who has 50 percent higher expected health expenditures than the low risk type at each age. We denote her expected lifecycle health expenditures conditional on survival as $E(m|\text{surv}, \text{high})$. Note that $E(m|\text{surv}, \text{low})$ and $E(m|\text{surv}, \text{high})$ would also represent the actuarially fair premiums of short-term spot contracts by age, for low and high risk types, respectively. $P_{30,\text{low}}$ ($P_{30,\text{high}}$) are the GLTHI premiums for a low (high) risk type who first enrolls in a GLTHI plan at age 30. Similarly, $P_{50,\text{low}}$ and $P_{50,\text{high}}$ are the premiums if the corresponding type joins a GLTHI plan at age 50.

Figure A3 has the following important features: First, in theory, premiums are stable over individuals’ lifecycles. Front-loading dampens age-driven premium increases via the legal requirement of building individual capital stocks through so-called old-age provisions (*Altersrückstellungen*). The capital stock is the cumulative difference between premiums and expected claims, plus returns on investment from the capital stock.^{A59} Second, premiums are higher for policyholders who sign a GLTHI contract later in their lives. As health expenditures increase with age, those who join later in life have fewer years to build up old-age provisions.^{A60} Third, because of the initial risk rating, high risk types pay higher premiums than low risk types throughout their lives. The following subsection discusses more institutional details about the premium calculation. It also discusses why, in reality,

^{A58}We assume constant health risks only for the purpose of illustrating the basic front-loading principle. Our empirical analysis below features evolving health risks, which is fundamental: First, front-loading can dampen the reclassification risk. Second, evolving health risks imply that individuals who start unhealthy may lapse their contract if their health improves over time. This introduces (downwards) reclassification risk even if premiums are constant within a given contract. Also, we must consider lapsation when calculating equilibrium premiums.

^{A59}In 2022, the average capital stock built through old-age provisions was €36,108 per policyholder (Association of German Private Healthcare Insurers, 2023)

^{A60}This is not necessarily true when health changes over time. With stochastic health, the initial premium may start to decrease at very high ages as, over time, the need to front-load for future health shocks decreases (see Section 6.1.)

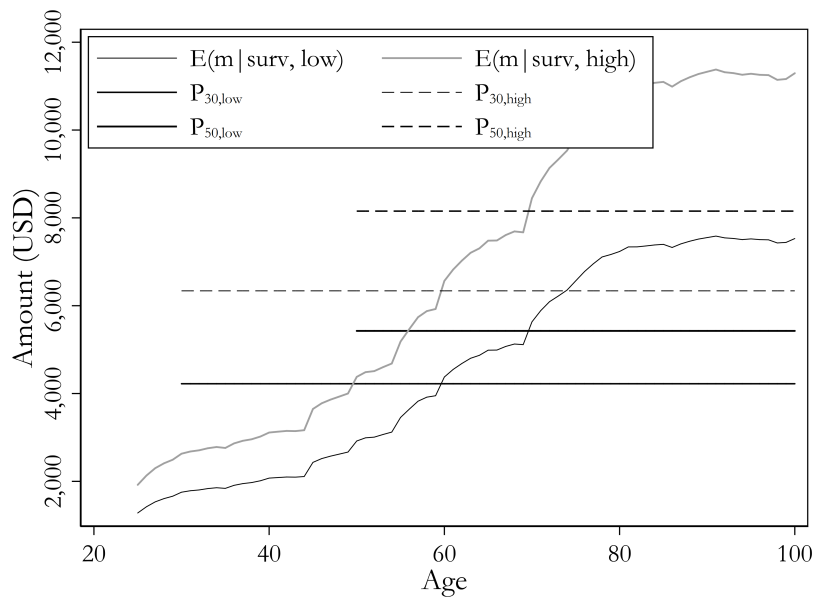


Figure A3: Illustration of GLTHI Lifecycle Premiums and Health Expenditures

Notes: The figure illustrates the principles of initial risk-rating and break-even for GLTHI plans over the life-cycle. It is based on Rosenbrock (2010) and Hofmann and Browne (2013). Source: Own calculations using German Claims Panel Data.

real premiums may not be flat over the lifecycle.

A2.2 Additional Details

The initial GLTHI premium is individually underwritten, but all subsequent premium changes over the lifecycle must be community rated.^{A61} Premiums consist of several components. The *Kalkulationsverordnung (KalV)* regulates the actuarial calculations. These calculations have to be approved by a federal financial regulatory agency (the *Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin*). The *KalV* specifies that premiums have to be a function of the expected per capita health care claims (*Kopfschäden*, which depend on the plan chosen, age, gender, and health risks),^{A62} the assumed guaranteed interest rate (*Rechnungszins*), the probability to lapse (*Stornowahrscheinlichkeit*), and the life expectancy (*Sterbewahrscheinlichkeit*).

One important and distinct characteristic of the GLTHI market is the legal obligation of insurers to

^{A61} The only exception is the Basic Plan (*Basistarif*). All insurers have to offer a Basic Plan. It follows the standardized GKV plan with the same essential benefits and actuarial values. For the Basic Plan, guaranteed issue exists for people above 55 and those who joined the GLTHI after 2009. The maximum premium is capped at the maximum GKV contribution (2021: €769,16 per month). The legislature mandated the Basic Plan to provide an “affordable” private option for GLTHI enrollees who cannot switch back to GKV, are uninsured, would have to pay excessive premiums, or would be denied coverage. However, the demand for the Basic Plan has been negligible; thus henceforth, we will abstain from it. In 2019, in the entire GLTHI, only 32,400 people, or 0.4 percent, were enrolled in the Basic Plan (Association of German Private Healthcare Insurers, 2020). In our data, only 1,006 enrollees chose the basic plan in 2010.

^{A62} Gender rating was common practice until December 21, 2012. After this date, for new contracts, all insurers in the European Union (EU) have to provide unisex premiums as the EU Court of Justice banned gender rating as discriminatory (Schmeiser et al., 2014)

build up *old-age provisions*, typically until age 60 of the policyholder. [Atal et al. \(2019\)](#) provide further details on this institutional element. The old-age provisions accumulated early in the policyholder's lifecycle serve as the capital stock to cover higher health expenditures later in the lifecycle.

Premiums are then calculated under the basic principle of a constant lifecycle premium. This premium is supposed to be sufficient to cover the policyholder's lifecycle health expenditures. Section 3.1 formally expresses this principle. Thus, in young ages, premiums exceed expected claims while in old ages, premiums fall short of expected claims—a phenomenon known as “front-loading” in long-term insurance contracts ([Hendel and Lizzeri, 2003](#); [Nell and Rosenbrock, 2007, 2009](#); [Fang and Kung, 2020](#)).^{A63}

While, theoretically, premiums are stable over individuals' lifecycles, in reality, nominal (and also real) premiums do increase. The main factors that trigger such “premium adjustments” (*Beitragsanpassungen*) are structural and unexpected changes in (i) life expectancy, (ii) health care consumption, (iii) health care prices, for example, due to new medical technology,^{A64} and (iv) and economic fundamentals. An example of (iv) was the shift of central banks to a super-low interest rate environment after the Great Recession in 2008; this shift implied a significant decrease in the returns to risk-free capital investment. Because GLTHI insurers (like life insurers) are heavily invested in the bond market, premium adjustments are a consequence of super-low interest rates.^{A65}

In some cases, premium adjustments are not only allowed, but *required* by the regulatory agency *BaFin*. This is to ensure financial stability within the regulatory framework of the *Versicherungsvertragsgesetz (VVG)*, the *Versicherungsaufsichtsgesetz (VAG)*, and the *KaIV*.^{A66} Most insurers have to follow the *Solvency II* reporting requirements. Each year, insurers have to check whether their underlying assumptions to calculate premiums and old age provision are still accurate. If they deviate by a certain amount, they have to adjust the premiums, which can result in two-digit premium increases, bad press, and lawsuits ([Krankenkassen-Zentrale \(KKZ\), 2020](#)).^{A67} However, average nom-

^{A63} Such front-loading creates a “lock-in” effect, in addition to the lock-in induced by guaranteed renewability ([Atal, 2019](#)). To strengthen consumer power and reduce this lock-in, the German legislature made a standardized portion of these old-age provisions portable across insurers for contracts signed after Jan 1, 2009; see [Atal et al. \(2019\)](#) for an evaluation of this reform. For existing contracts, [Atal et al. \(2019\)](#) do not find a significant impact on the likelihood to switch insurers.

^{A64} The Health Care Reform 2000 (*GKV-Gesundheitsreformgesetz 2000*) introduced a mandatory 10 percent premium surcharge up to age 60 to dampen structural increases in health care spending due to medical progress. This surcharge only applies to GLTHI contracts signed after January 1, 2000 (see article 14 of *GKV-Gesundheitsreformgesetz (2000)*).

^{A65} The *KaIV* has traditionally capped the assumed return on equity—the “guaranteed interest rate” (*Rechnungszins*)—at 3.5 percent for the premium calculation. This was the case for five decades. However, in 2016 for the first time, the average net return on investment has dropped below 3.5 percent, which is why the *German Actuary Association* has issued a new guideline to calculate the new insurer-specific “maximum allowed interest rate” (*Höchstrechnungszins*), see [Deutsche Aktuarvereinigung \(DAV\) \(2019\)](#).

^{A66} Effective January 1, 2016 the *KaIV* has been replaced by the *Krankenversicherungsaufsichtsverordnung (KVAV)*.

^{A67} All premium adjustments have to be legally checked and approved by 16 independent actuaries who are appointed by the *BaFin*. However, some plaintiffs in lawsuits argue that some of these actuaries would not be sufficiently independent. Other reasons of courts to declare a premium increase as “not justified” were insufficient explanations by the insurers or

inal premium increases have been moderate in an international comparison. According to data by the Association of German Private Healthcare Insurers (*Verband der Privaten Krankenversicherung*) total revenues from premiums increased annually between 0 and 4.5 percent from 2011 to 2020 ([Bundesministerium für Gesundheit, 2024](#)); the industry reports average annual premium increases of 2.6% from 2012-2022 ([Bahnsen and Wild, 2021](#)). Most important for our analysis is that, after the initial risk rating, premium adjustments do not depend on policyholders' evolving health status.

A3 Proof for Lemma 1

We prove the lemma through induction. With $b_T = 0$, we first show that Equation (3) holds for period $T - 1$, i.e. $b_{T-1}(\xi_T | \Xi_{T-1})$. Then, we show that Equation (3) holds for period t , given it holds for $t + 1$. First, note that for the securities to replicate the premium path of the GLTHI contract, the following equations must hold

$$-\tilde{P}_t(\Xi_t) = -E(m | \xi_t) - \sum_{\xi_{t+1} \in \mathcal{Z}} \delta \pi(\xi_{t+1} | \xi_t) b_t(\xi_{t+1} | \Xi_t) + b_{t-1}(\xi_t | \Xi_{t-1}) \quad \text{for } t \in \{1, 2, \dots, T\} \quad (8)$$

with $b_0 = 0$ and $b_T = 0$. The previous expressions ensure that the addition of the premium of the short-term contract and the net proceeds from the Arrow securities equate the premium paid in the GLTHI contract after history Ξ_t .

Terminal Period. In the terminal period, the premium of a GLTHI contract in the spot market is equal to the premium of a short term contract: $P(\xi_T) = E(m_T | \xi_T)$. The premium paid under the GLTHI contract is therefore $\tilde{P}_T(\Xi_T) = \min \{P_T(\xi_T), \tilde{P}_{T-1}(\Xi_{T-1})\}$. There are two relevant cases:

1. $P_T(\xi_T) \leq \tilde{P}_{T-1}(\Xi_{T-1})$ (*lapse in T*). From equation (8), we get:

$$b_{T-1}(\xi_T | \Xi_{T-1}) = E(m_T | \xi_T) - \tilde{P}_T(\Xi_T) = P_T(\xi_T) - P_T(\xi_T) = 0$$

2. $P_T(\xi_T) > \tilde{P}_{T-1}(\Xi_{T-1})$ (*no lapse in T*). From equation (8), we get:

$$b_{T-1}(\xi_T | \Xi_{T-1}) = E(m_T | \xi_T) - \tilde{P}_T(\Xi_T) = E(m_T | \xi_T) - \tilde{P}_{T-1}(\Xi_{T-1})$$

Therefore equation (3) holds for $T - 1$.

deliberate initial underpricing in the first year to attract enrollees ([Krankenkassen-Zentrale \(KKZ\), 2020](#)).

Period $t < T - 1$. We now consider a period $t < T - 1$. First, note that $\mathbb{1}(P_{t+2}(\xi_{t+2}) > \tilde{P}_{t+1}(\Xi_{t+1})) = 1 \iff b_{t+1}(\xi_{t+2} | \Xi_{t+1}) > 0$, and therefore,

$$\pi(\xi_{t+2} | \xi_{t+1}) b_{t+1}(\xi_{t+2} | \Xi_{t+1}) = \pi(\xi_{t+2} | \xi_{t+1}) b_{t+1}(\xi_{t+2} | \Xi_{t+1}) \mathbb{1}(P_{t+2}(\xi_{t+2}) > \tilde{P}_{t+1}(\Xi_{t+1}))$$

Given that equation (3) holds for period $t + 1$,

$$\begin{aligned} \pi(\xi_{t+2} | \xi_{t+1}) b_{t+1}(\xi_{t+2} | \Xi_{t+1}) &= (E(m_{t+2} | \xi_{t+2}) - \tilde{P}_{t+1}(\Xi_{t+1})) \pi(\xi_{t+2} | \xi_{t+1}) \mathbb{1}(P_{t+2}(\xi_{t+2}) > \tilde{P}_{t+1}(\Xi_{t+1})) \\ &+ \left(\sum_{\tau > t+2}^T \sum_{z \in \mathcal{Z}} \delta^{\tau-(t+2)} (E(m_\tau | \xi_\tau) - \tilde{P}_{t+1}(\Xi_{t+1})) \times q_\tau(z | \xi_{t+2}, \mathbf{P}_{t+3}^\tau, \tilde{P}_{t+1}(\Xi_{t+1})) \right) \pi(\xi_{t+2} | \xi_{t+1}) \mathbb{1}(P_{t+2}(\xi_{t+2}) > \tilde{P}_{t+1}(\Xi_{t+1})) \end{aligned}$$

By definition, $q_\tau(\xi_\tau | \xi_{t+1}, \mathbf{P}_{t+2}^\tau, \tilde{P}_t(\Xi_t)) = \prod_{j=t+2}^\tau [\mathbb{1}(P_j(\xi_j) > \tilde{P}_t(\Xi_t)) \pi(\xi_{j+1} | \xi_j)]$. Therefore, we can re-write the expression above as:

$$\begin{aligned} \pi(\xi_{t+2} | \Xi_{t+1}) b_{t+1}(\xi_{t+2} | \Xi_{t+1}) &= (E(m_{t+2} | \xi_{t+2}) - \tilde{P}_{t+1}(\Xi_{t+1})) \times q_{t+2}(\xi_{t+2} | \xi_{t+1}, \mathbf{P}_{t+2}^{t+2}, \tilde{P}_{t+1}(\Xi_{t+1})) \\ &+ \sum_{\tau > t+2}^T \sum_{z \in \mathcal{Z}} \delta^{\tau-(t+2)} (E(m_\tau | \xi_\tau) - \tilde{P}_{t+1}(\Xi_{t+1})) \times q_\tau(z | \xi_{t+1}, \mathbf{P}_{t+2}^\tau, \tilde{P}_{t+1}(\Xi_{t+1})) \end{aligned}$$

Summing across ξ_{t+2} and multiplying by δ , we get:

$$\delta \sum_{\xi_{t+2}} \pi(\xi_{t+2} | \Xi_{t+1}) b(\xi_{t+2} | \Xi_{t+1}) = \sum_{\tau > t+1}^T \sum_{z \in \mathcal{Z}} \delta^{\tau-(t+1)} [E(m_\tau | \xi_\tau) - \tilde{P}_{t+1}(\Xi_{t+1})] \times q_\tau(z | \xi_{t+1}, \mathbf{P}_{t+2}^\tau, \tilde{P}_{t+1}(\Xi_{t+1}))$$

We now consider the two relevant cases.

1. $P_{t+1}(\xi_{t+1}) \leq \tilde{P}_t(\Xi_t)$ (*lapse in $t + 1$*). We have $\tilde{P}_{t+1}(\Xi_{t+1}) = P_{t+1}(\xi_{t+1})$. From equation (8) we get:

$$\begin{aligned} b_t(\xi_{t+1} | \Xi_t) &= E(m_{t+1} | \xi_{t+1}) - \tilde{P}_{t+1}(\Xi_{t+1}) + \delta \sum_{\xi_{t+2}} \pi(\xi_{t+2} | \Xi_{t+1}) b_{t+1}(\xi_{t+2} | \Xi_{t+1}) \\ &= E(m_{t+1} | \xi_{t+1}) - \tilde{P}_{t+1}(\Xi_{t+1}) + \sum_{\tau > t+1}^T \sum_{z \in \mathcal{Z}} \delta^{\tau-(t+1)} [E(m_\tau | \xi_\tau) - \tilde{P}_{t+1}(\Xi_{t+1})] \times q_\tau(z | \xi_{t+1}, \mathbf{P}_{t+2}^\tau, \tilde{P}_{t+1}(\Xi_{t+1})) \\ &= E(m_{t+1} | \xi_{t+1}) - P_{t+1}(\xi_{t+1}) + \sum_{\tau > t+1}^T \sum_{z \in \mathcal{Z}} \delta^{\tau-(t+1)} [E(m_\tau | \xi_\tau) - P_{t+1}(\xi_{t+1})] \times q_\tau(z | \xi_{t+1}, \mathbf{P}_{t+2}^\tau, P_{t+1}(\xi_{t+1})) \\ &= 0 \end{aligned}$$

where the last step follows from the insurer's zero-profit condition.

2. $P_{t+1}(\xi_{t+1}) > \tilde{P}_t(\Xi_t)$ (no lapsation in $t + 1$). We have $\tilde{P}_{t+1}(\Xi_{t+1}) = \tilde{P}_t(\Xi_t)$. From equation (8) we get:

$$\begin{aligned} b_t(\xi_{t+1} | \Xi_t) &= E(m_{t+1} | \xi_{t+1}) - \tilde{P}_{t+1}(\Xi_{t+1}) + \\ &\sum_{\tau > t+1}^T \sum_{z \in \mathcal{Z}} \delta^{\tau-(t+1)} [E(m_\tau | \xi_\tau) - \tilde{P}_{t+1}(\Xi_{t+1})] \times q_\tau(z | \xi_{t+1}, \mathbf{P}_{t+2}^\tau, \tilde{P}_{t+1}(\Xi_{t+1})) \\ &= E(m_{t+1} | \xi_{t+1}) - \tilde{P}_t(\Xi_t) + \sum_{\tau > t+1}^T \sum_{z \in \mathcal{Z}} \delta^{\tau-(t+1)} [E(m_\tau | \xi_\tau) - \tilde{P}_t(\Xi_t)] \times q_\tau(z | \xi_{t+1}, \mathbf{P}_{t+2}^\tau, \tilde{P}_t(\Xi_t)) \end{aligned}$$

Therefore, equation (3) also holds for period t .

Then the equation holds for all $t \in \{1, 2, \dots, T\}$.

A3.1 Characterizing Optimal Contracts with Arrow Securities

This section shows how to implement the optimal dynamic long-term contracts as in [Chili et al. \(2024\)](#) with one-sided commitment by trading state-contingent one-period Arrow securities.

First, we define the individual's consumption when only short-term contracts are available $c_t(\xi_t)$:

$$c_t(\xi_t) = y_t - E(m_t | \xi_t).$$

Equation (2) provides the individual's long-term guaranteed consumption offered by a competitive insurer. We define the guaranteed consumption from the previous period, $\tilde{c}_t$:

$$\tilde{c}_t(\Xi_t) = \max \{ \tilde{c}_{t-1}(\Xi_{t-1}), \bar{c}_t(\xi_t) \}$$

We aim to find the quantities of Arrow securities that would replicate the consumption path under the optimal contract when combined with short-term contracts. In particular, it must be true that

$$\tilde{c}_t(\Xi_t) = y_t - E(m_t | \xi_t) - \sum_{\xi_{t+1} \in \mathcal{Z}} \delta \pi(\xi_{t+1} | \xi_t) b_t(\xi_{t+1} | \Xi_t) + b_{t-1}(\xi_t | \Xi_{t-1}) \quad \text{for } t \in \{1, 2, \dots, T\}$$

with $b_0 = 0$ and $b_T = 0$

Lemma 2 *The consumption path under the optimal dynamic contract can be replicated by purchasing short-term insurance contracts supplemented by Arrow securities. The quantity of Arrow securities bought after*

history Ξ_t that pay one dollar in state ξ_{t+1} are equal to

$$b_t(\xi_{t+1} \mid \Xi_t) = \begin{cases} 0 & \text{if } \bar{c}_{t+1}(\xi_{t+1}) > \tilde{c}_t(\Xi_t) \\ (\tilde{c}_t(\Xi_t) - c_{t+1}(\xi_{t+1})) + \\ \sum_{\tau > t+1}^T \sum_{z \in \mathcal{Z}} \delta^{\tau-(t+1)} [\tilde{c}_t(\Xi_t) - c_\tau(\xi_\tau)] q_\tau(z \mid \xi_{t+1}, \mathbf{C}_{t+2}^\tau, \tilde{c}_t(\Xi_t)) & \text{otherwise} \end{cases} \quad (9)$$

where $q_\tau(z \mid \xi_{t+1}, \mathbf{C}_{t+2}^\tau, \tilde{c}_t(\Xi_t))$ is the probability that (i) $\xi_\tau = z$, and (ii) the enrollee does not lapse (or die) between periods $t+1$ and τ , given the subsequent equilibrium consumption $\mathbf{C}_{t+2}^\tau$ and the guaranteed-renewable consumption $\tilde{c}_t(\Xi_t)$.

A4 Descriptive Statistics for GLTHI Dataset and SOEP Dataset

Table A1: Summary Statistics: German Claims Panel Data

	Mean	SD	Min	Max	N
Socio-Demographics					
Age (in years)	45.5	11.4	25.0	99.0	1,867,465
Female	0.276	0.447	0.0	1.0	1,867,465
Policyholder since (years)	6.5	5.0	1.0	40.0	1,867,465
Client since (years)	12.8	11.0	1.0	86.0	1,867,465
Employee	0.336	0.473	0.0	1.0	1,867,465
Self-Employed	0.486	0.500	0.0	1.0	1,867,465
Civil Servant	0.132	0.338	0.0	1.0	1,867,465
Health Risk Penalty	0.358	0.480	0.0	1.0	1,867,465
Pre-Existing Condition Exempt	0.016	0.126	0.0	1.0	1,867,465
Health Plan Parameters					
TOP Plan	0.377	0.485	0.0	1.0	1,867,465
PLUS Plan	0.338	0.473	0.0	1.0	1,867,465
ECO Plan	0.285	0.451	0.0	1.0	1,867,465
Annual premium (USD)	4,749	2,157	0	33,037	1,867,318
Annual risk penalty (USD)	157	453	0	21,752	1,867,465
Deductible(USD)	675	659	0	3,224	1,867,465
Total Claims (USD)	3,289	8,577	0	2,345,126	1,867,465

Source: German Claims Panel Data. *Policyholder since* is the number of years since the client has enrolled in the current plan; *Client since* is the number of years since the client joined the company. *Employee* and *Self-Employed* are dummies for the policyholders' current occupation. *Health Risk Penalty* is a dummy that is one if the initial underwriting led to a health-related risk penalty on top of the factors age, gender, and type of plan; *Pre-Existing Conditions Exempt* is a dummy that is one if the initial underwriting led to exclusions of pre-existing conditions. The mutually exclusive dummies *TOP Plan*, *PLUS Plan* and *ECO Plan* capture the generosity of the plan. *Annual premium* is the annual premium, and *Annual Risk Penalty* is the amount of the health risk penalty charged. *Deductible* is the deductible and *Total Claims* the sum of all claims in a calendar year. See Section 4.1 for further details.

Table A2: Summary Statistics: German Socio-Economic Panel Study

	Mean	SD	Min	Max	N
Socio-Demographics					
Female	0.5217	0.4995	0	1	530,228
Age	46.9119	17.4922	17	105	530,228
No degree yet	0.058	0.2338	0	1	530,228
Dropout of high school	0.0378	0.1908	0	1	530,228
Degree after 8/9 years of schooling (Ed 8)	0.3619	0.4805	0	1	530,228
Degree after 10 years of schooling (Ed 10)	0.2737	0.4459	0	1	530,228
Degree after 13 years of schooling (Ed 13)	0.1746	0.3796	0	1	530,228
Employment					
Civil servant	0.0393	0.1943	0	1	530,228
Self-employed	0.0624	0.2419	0	1	530,228
White collar	0.2736	0.4458	0	1	530,228
Full-time employed	0.4152	0.4928	0	1	530,228
Part-time employed	0.1402	0.3471	0	1	530,228
Income Measures in 2016 USD					
Monthly gross wage	2,940	2,506	0	215,093	310,460
Monthly net wage	1,921	1,527	0	134,511.5	310,460
Individual annual total income	20,361	24,434	0	2,580,000	530,228
Equivalized post-tax post-transfer annual income	26,433	18,731	0	2,155,394	530,228
Insurance and Utilization					
Hospital nights in past calendar year	1.6652	8.3794	0	365	530,228
Doctor visits in past 3 months	2.4941	4.1436	0	99	461,971
Privately insured	1	0	1	1	57,558

Source: SOEP (2018), the long version from 1984 to 2016. Whenever the number of person-year observations is less than 530,228 the question was not asked in all years from 1984 to 2016. For example, *Doctor visits in past 3 months* has only been routinely asked since 1995. *Privately insured* indicates that 57,558/530,228=10.8% of all observations are by people who are insured on the GLTHI market. All income measures have been consistently generated and cleaned by the SOEP team; e.g., *Monthly gross wage* is labeled *labgro* and *Monthly net wage* is labeled *labnet* in SOEP (2018). See Section 4.2 for a detailed discussion of the variables.

A5 Risk Classification: Further Details and Robustness Checks

A5.1 Efficient Classification

According to the actuarial science literature (Finger, 2001), an efficient risk classification system has two properties: *homogeneity*—meaning that individuals in the same risk category have similar risk and *separation*—meaning that the categories have sufficiently different expected claims to justify distinct categories.^{A68}

^{A68}For instance, in Figure 1, it is easy to see that equally-sized categories are unlikely to be optimal as they would assign similar individuals in terms of λ^* into different categories in the left tail of the distribution, failing the *separation* principle.

We define a *risk classification* as a surjective function $f_K : \mathbb{R}_+^n \rightarrow \{\lambda \in \mathbb{Z} : 1 \leq \lambda \leq K\}$, where $\mathbb{R}_+^n$ is the state space (i.e. λ_t^* and its $n - 1$ lags, a vector we denote $\{\lambda_{t-v}^*\}_{v=0}^n$). Denote this classification function $\lambda_t = f_K \left(\{\lambda_{t-v}^*\}_{v=0}^n \right)$, where $\lambda_t \in \{1, \dots, K\}$ is the risk category assigned to a person with those ACG scores. The *efficient* risk classification f_K maximizes the “structure variance”

$$SV(f_K) = \text{Var}(m_t) - \sum_{k=1}^K \Pr(\lambda_t = k) \text{Var}(m_t \mid \lambda_t = k), \quad (10)$$

where m_t is individual annual health expenditures (Finger, 2001). The structure variance $SV(f_K)$ is thus the total variance less the weighted sum of within-class variances of health expenditures. Put differently, the efficient classification maximizes the variance of mean expenditure across groups. Applying the law of total variance to both terms in Equation (10), we can write the structure variance as:^{A69}

$$SV(f_K) = \text{Var}(\mathbb{E}(m_t \mid \{\lambda_{t-v}^*\}_{v=0}^n)) - \sum_{k=1}^K \Pr(\lambda_t = k) \text{Var}(\mathbb{E}(m_t \mid \{\lambda_{t-v}^*\}_{v=0}^n) \mid \lambda_t = k). \quad (11)$$

Note that the first term in Equation (11) is independent of the classification (as it is independent of the classes λ_t). Thus for a given K , finding the efficient classification system is equivalent to finding the classes λ_t that *minimize* the total heterogeneity in expected expenditure within risk classes: $\sum_{k=1}^K \Pr(\lambda_t = k) \text{Var}(\mathbb{E}(m_t \mid \Lambda_t^*(n)) \mid \lambda_t = k)$.

Three additional things are worth noting about Equation (11). First, only mean expenditures conditional on ACG scores $\mathbb{E}(m_t \mid \{\lambda_{t-v}^*\}_{v=0}^n)$ matter for the classification system, whereas the dispersion of m_t around this mean is inconsequential. Second, minimizing heterogeneity within classes is incidentally what the k-means clustering method does (Lloyd, 1982; Athey and Imbens, 2019). Thus, we will apply k-means clustering of $\mathbb{E}(m_t \mid \{\lambda_{t-v}^*\}_{v=0}^n)$ to determine the efficient classification system. Third, this implies that the efficient classification also maximizes the coefficient of determination (R^2) in a regression of expenditures on risk class indicators (Kriegel et al., 2017).

A5.2 Further Details

Smoothing. To get accurate predictions for expected claims along the entire distribution of risk, including the tails, we use cubic regression splines in the estimation of $\mathbb{E}(m_t \mid \{\lambda_{t-v}^*\}_{v=0}^n)$. Figure

In addition, it would assign individuals with substantial λ^* differences into identical categories in the right tail of the distribution, failing the *homogeneity* principle.

^{A69}The law of total variance implies $\text{Var}(m_t) = \mathbb{E}(\text{Var}(m_t \mid \Lambda_t^*(n))) + \text{Var}(\mathbb{E}(m_t \mid \Lambda_t^*(n)))$ and $\text{Var}(m_t \mid \lambda_t = k) = \mathbb{E}(\text{Var}(m_t \mid \Lambda_t^*(n)) \mid \lambda_t = k) + \text{Var}(\mathbb{E}(m_t \mid \Lambda_t^*(n)) \mid \lambda_t = k)$.

A4 provides a comparison of mean expenditure by $\{\lambda_{t-v}^*\}_{v=0}^n$ before and after smoothing for $n = 2$.

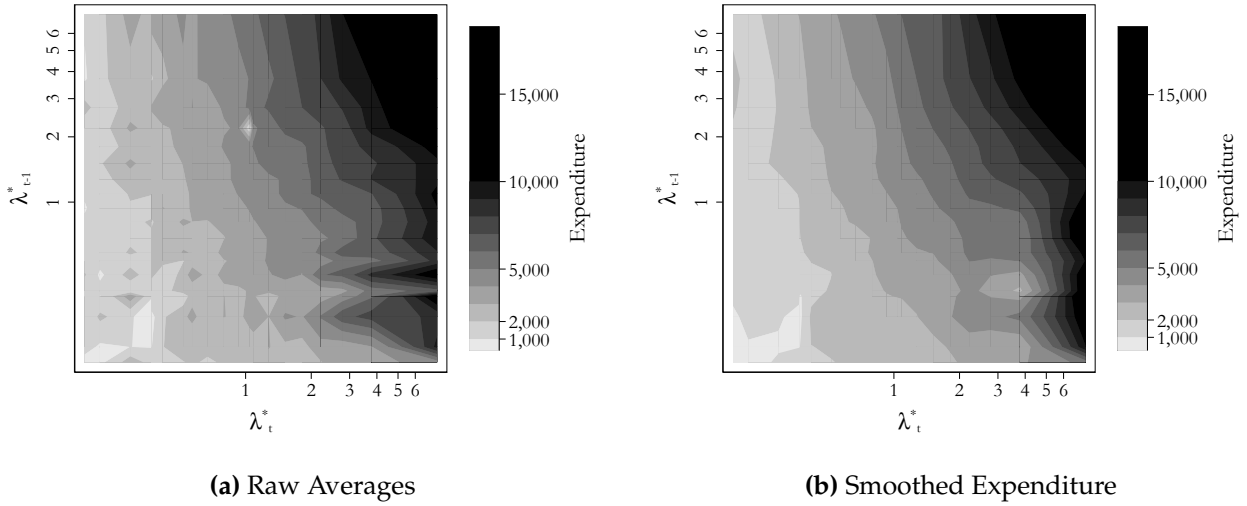


Figure A4: Mean Expenditure by $\{\lambda_{t-v}^*\}_{v=0}^n$.

Note: The left figure is based on average expenditure within each of 400 cells (ventiles in λ_t^* and λ_{t-1}^*). The right figure uses predicted values from a cubic spline regression. Source: German Claims Panel Data.

Stochastic Dominance. In their characterization of the optimal contract, [Ghili et al. \(2024\)](#) invoke an assumption of stochastic dominance. It requires that transition rates between risk categories—which are represented by the cumulative distribution function $F(\lambda_{t+1} | \lambda_t)$ —satisfy first-order stochastic dominance in the following sense: if $\lambda'_t > \lambda_t$, then $F(\lambda_{t+1} | \lambda'_t) \succ_{FSD} F(\lambda_{t+1} | \lambda_t)$. Figure **A5** shows that this property holds for all pairwise combinations of (λ_t, λ'_t) such that $\lambda'_t > \lambda_t$.

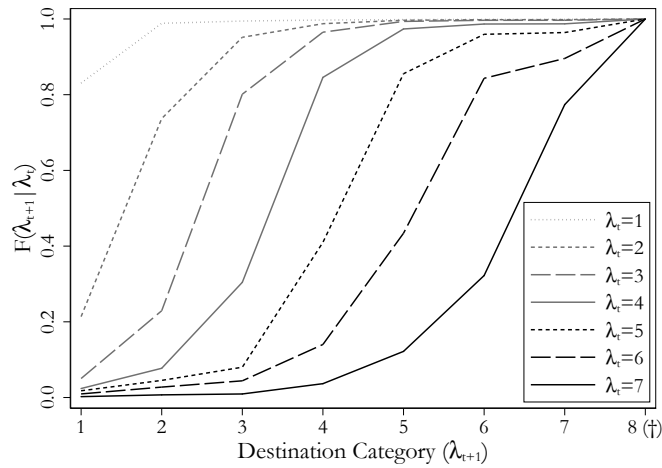


Figure A5: Stochastic Dominance.

Note: The figure shows the cumulative distribution function $F(\lambda_{t+1} | \lambda_t)$ for all pairwise combinations of (λ_t, λ'_t) such that $\lambda'_t > \lambda_t$. Source: German Claims Panel Data.

Predicted Expenditures. Figure **A6** shows the estimated mean expenditure by age for each risk category.

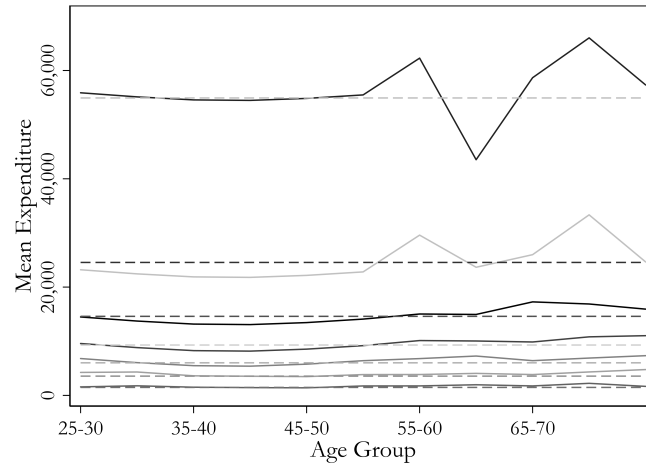


Figure A6: Predicted Health Expenditure

Note: Solid curves represent mean expenditure by age for each risk category λ_t , estimated according to Equation (5) in Section 5.2. The dashed lines represent the corresponding predictions assuming expenditure does not depend on age. Source: German Claims Panel Data.

Transition Matrices. Table A3 and A4 show risk category transition matrices for those aged 25-54 and 55+, respectively.

Table A3: λ Risk Category Transitions by Age Group—Ages 25–54

Age	λ_t	λ_{t+1}							
		1	2	3	4	5	6	7	8 (†)
25-29	1	0.8907	0.1024	0.0047	0.0011	0.0004	0.0003	0.0001	0.0004
	2	0.3197	0.4257	0.2020	0.0432	0.0077	0.0011	0.0003	0.0003
	3	0.1242	0.2829	0.4104	0.1404	0.0378	0.0043	0.0000	0.0000
	4	0.0892	0.1688	0.2484	0.3917	0.0860	0.0159	0.0000	0.0000
	5	0.0938	0.1250	0.0625	0.3750	0.2917	0.0521	0.0000	0.0000
	6	0.0909	0.0000	0.0455	0.2273	0.3182	0.3182	0.0000	0.0000
	7	0.0000	0.0000	0.0002	0.0045	0.0240	0.1447	0.7619	0.0647
30-34	1	0.8767	0.1145	0.0055	0.0018	0.0009	0.0002	0.0001	0.0003
	2	0.3212	0.4347	0.1909	0.0438	0.0080	0.0006	0.0001	0.0007
	3	0.1241	0.3015	0.4080	0.1409	0.0229	0.0016	0.0000	0.0011
	4	0.1039	0.1640	0.2407	0.3739	0.1032	0.0115	0.0007	0.0021
	5	0.0734	0.0911	0.0506	0.2911	0.3747	0.1089	0.0025	0.0076
	6	0.0422	0.0438	0.0529	0.1678	0.3628	0.2450	0.0525	0.0329
	7	0.0128	0.0115	0.0083	0.0574	0.1545	0.1663	0.4524	0.1368
35-39	1	0.8427	0.1480	0.0055	0.0022	0.0009	0.0002	0.0001	0.0004
	2	0.2798	0.4635	0.2113	0.0360	0.0076	0.0013	0.0000	0.0005
	3	0.1177	0.2379	0.4850	0.1288	0.0275	0.0028	0.0001	0.0002
	4	0.0719	0.0967	0.3055	0.4085	0.0999	0.0158	0.0003	0.0014
	5	0.0743	0.0493	0.0691	0.3402	0.3629	0.0958	0.0039	0.0045
	6	0.0415	0.0331	0.0340	0.1180	0.2958	0.4009	0.0455	0.0312
	7	0.0127	0.0088	0.0054	0.0409	0.1276	0.2757	0.3975	0.1313
40-44	1	0.8514	0.1392	0.0050	0.0024	0.0010	0.0003	0.0001	0.0006
	2	0.2862	0.4666	0.2050	0.0329	0.0075	0.0014	0.0001	0.0003
	3	0.1137	0.2229	0.5134	0.1225	0.0241	0.0022	0.0001	0.0011
	4	0.0790	0.0769	0.2936	0.4213	0.1113	0.0157	0.0003	0.0018
	5	0.0640	0.0392	0.0759	0.3281	0.3763	0.1055	0.0038	0.0072
	6	0.0295	0.0382	0.0342	0.1605	0.2773	0.3613	0.0539	0.0450
	7	0.0081	0.0091	0.0049	0.0502	0.1079	0.2240	0.4247	0.1710
45-49	1	0.8148	0.1736	0.0059	0.0028	0.0012	0.0006	0.0002	0.0009
	2	0.2267	0.5059	0.2229	0.0329	0.0093	0.0013	0.0001	0.0010
	3	0.0653	0.2027	0.5708	0.1309	0.0258	0.0031	0.0001	0.0012
	4	0.0427	0.0712	0.2877	0.4655	0.1153	0.0140	0.0005	0.0029
	5	0.0303	0.0438	0.0475	0.3570	0.3964	0.1101	0.0058	0.0090
	6	0.0153	0.0266	0.0211	0.1118	0.2919	0.4163	0.0607	0.0563
	7	0.0038	0.0057	0.0027	0.0314	0.1021	0.2321	0.4298	0.1923
50-54	1	0.8117	0.1740	0.0056	0.0035	0.0020	0.0008	0.0004	0.0020
	2	0.2283	0.4979	0.2228	0.0377	0.0101	0.0016	0.0002	0.0015
	3	0.0602	0.1799	0.5727	0.1509	0.0317	0.0027	0.0001	0.0018
	4	0.0398	0.0648	0.2660	0.4930	0.1160	0.0155	0.0007	0.0041
	5	0.0274	0.0387	0.0426	0.3666	0.3866	0.1182	0.0075	0.0124
	6	0.0130	0.0222	0.0179	0.1084	0.2688	0.4220	0.0746	0.0732
	7	0.0028	0.0042	0.0020	0.0265	0.0819	0.2049	0.4600	0.2176

Note: The table shows transition probabilities across risk categories by age group. Source: German Claims Panel Data.

Table A4: λ Risk Category Transitions by Age Group—Ages 55+

Age	λ_t	λ_{t+1}							
		1	2	3	4	5	6	7	8 (†)
55-59	1	0.7261	0.2537	0.0101	0.0037	0.0020	0.0013	0.0004	0.0027
	2	0.0932	0.6432	0.2123	0.0357	0.0110	0.0018	0.0004	0.0025
	3	0.0002	0.1739	0.6167	0.1690	0.0335	0.0044	0.0001	0.0024
	4	0.0001	0.0637	0.2426	0.5404	0.1287	0.0180	0.0007	0.0058
	5	0.0001	0.0356	0.0363	0.3758	0.4009	0.1282	0.0069	0.0163
	6	0.0000	0.0195	0.0145	0.1061	0.2662	0.4370	0.0650	0.0917
	7	0.0000	0.0037	0.0016	0.0260	0.0813	0.2126	0.4016	0.2732
60-64	1	0.7558	0.2147	0.0145	0.0044	0.0042	0.0019	0.0011	0.0033
	2	0.1023	0.6414	0.1981	0.0387	0.0120	0.0031	0.0004	0.0040
	3	0.0002	0.1612	0.6076	0.1836	0.0394	0.0053	0.0001	0.0028
	4	0.0001	0.0555	0.2243	0.5507	0.1419	0.0204	0.0008	0.0063
	5	0.0001	0.0292	0.0317	0.3610	0.4168	0.1370	0.0075	0.0168
	6	0.0000	0.0153	0.0122	0.0980	0.2660	0.4489	0.0686	0.0910
	7	0.0000	0.0028	0.0013	0.0235	0.0794	0.2136	0.4143	0.2651
65-69	1	0.3707	0.5949	0.0172	0.0076	0.0030	0.0015	0.0009	0.0042
	2	0.0624	0.6492	0.2407	0.0352	0.0065	0.0012	0.0004	0.0045
	3	0.0008	0.1058	0.6561	0.2082	0.0223	0.0013	0.0000	0.0056
	4	0.0002	0.0335	0.2013	0.6242	0.1261	0.0052	0.0005	0.0090
	5	0.0000	0.0128	0.0159	0.3546	0.4985	0.0763	0.0019	0.0400
	6	0.0000	0.0000	0.0107	0.0551	0.4067	0.3517	0.0195	0.1563
	7	0.0006	0.0066	0.0029	0.0264	0.0553	0.1690	0.5289	0.2103
70-74	1	0.3848	0.5793	0.0225	0.0060	0.0011	0.0003	0.0014	0.0048
	2	0.0070	0.6771	0.2554	0.0406	0.0105	0.0012	0.0000	0.0082
	3	0.0001	0.0810	0.6277	0.2599	0.0230	0.0014	0.0001	0.0068
	4	0.0002	0.0115	0.1625	0.6579	0.1404	0.0080	0.0002	0.0195
	5	0.0000	0.0015	0.0184	0.2829	0.5654	0.0736	0.0010	0.0572
	6	0.0000	0.0000	0.0000	0.0327	0.3039	0.4052	0.0065	0.2516
	7	0.0005	0.0056	0.0033	0.0184	0.0172	0.0263	0.7192	0.2094
75+	1	0.1770	0.5900	0.0442	0.0995	0.0598	0.0063	0.0083	0.0150
	2	0.0006	0.6237	0.2903	0.0471	0.0094	0.0012	0.0000	0.0277
	3	0.0000	0.0525	0.5876	0.2988	0.0254	0.0012	0.0000	0.0344
	4	0.0000	0.0029	0.1012	0.6668	0.1623	0.0055	0.0008	0.0605
	5	0.0000	0.0000	0.0060	0.2262	0.5581	0.0837	0.0028	0.1232
	6	0.0000	0.0000	0.0019	0.0206	0.3127	0.4064	0.0225	0.2360
	7	0.0000	0.0000	0.0000	0.0000	0.1111	0.1481	0.4630	0.2778

Notes: The table shows transition probabilities across risk categories by age group. Source: German Claims Panel Data.

A5.3 Robustness Checks

Winsorizing. First, we analyse the extent to which results are driven by outliers in m_{it} . It is of course desirable that outliers are considered in the classification, given their disproportionate contributions to means and variances; however, if the performance of the classification were widely different when they are not considered, it would cast doubt on how well the scheme performs with regard to less extreme risks. Therefore, we compared the performance of different classification schemes after the top percentile of expenditure had been winsorized. Results are provided in Figure A7. As

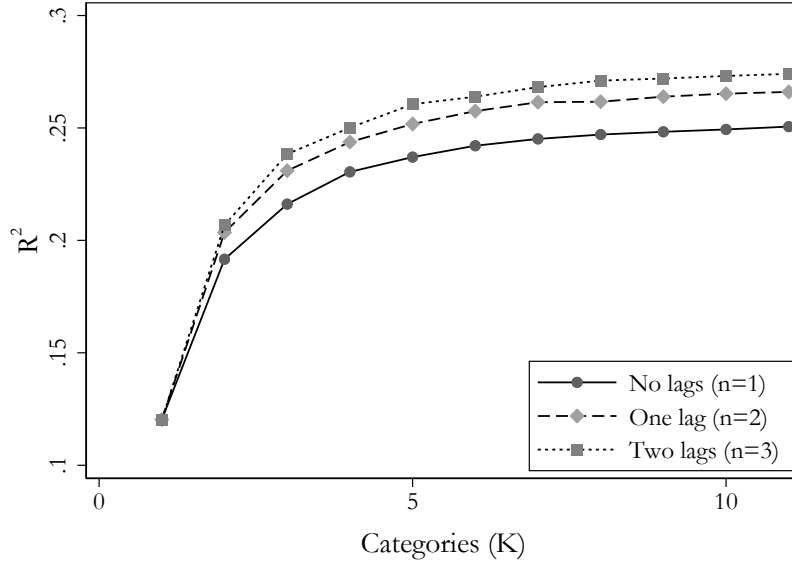


Figure A7: Performance of Alternative Risk Classifications: Winsorized Expenditure.

Note: The figure displays *unadjusted* R^2 of a regression of expenditures on risk category indicators. All results are robust to using the *adjusted* R^2 . Each specification includes 21 age-gender fixed effects, year fixed effects and 79 plan fixed effects. Source: Own Calculations using German Claims Panel Data.

expected, the topcoding of outliers improves the predictive power of all schemes; however, their relative performance is unaffected by this change.

Lags of classes. Second, we compare two different ways of including a longer history of claims. Instead of expanding on the information set Λ_t before discretizing, we consider an alternative based on $\Lambda_t^* = \lambda_t^*$ but with the predictive power of the classification scheme interacted with its lags (i.e. a classification based on K^2 classes). Figure A8 provides the results, comparing the two alternatives $q = 0$ and $q = 1$ from above. In an interacted version, the classification uses $q = 0$ but is interacted with its lags in the regressions (leading effectively to K^2 classes). This interacted alternative has similar, even better, predictive power than $q = 1$. However, the variant with $q = 1$ achieves similar performance with a much smaller number of classes.

Sample selection. The results in Figure 2 are based on a sample of individuals who are observed over 4 years, since three lags are needed in Λ_{it}^* . In Figure A9 we check how robust the finding is to varying the observation window for sample selection. Sample 1 requires only that m_i and λ_i^* are observed. Sample 2 requires also that λ_{t-1}^* is observed, and sample 3 requires in addition that λ_{t-2}^* is observed. Figure A9 shows the result and that the predictive performance is sensitive to the sample used; however, the relative performance between schemes is the same regardless of the sample.

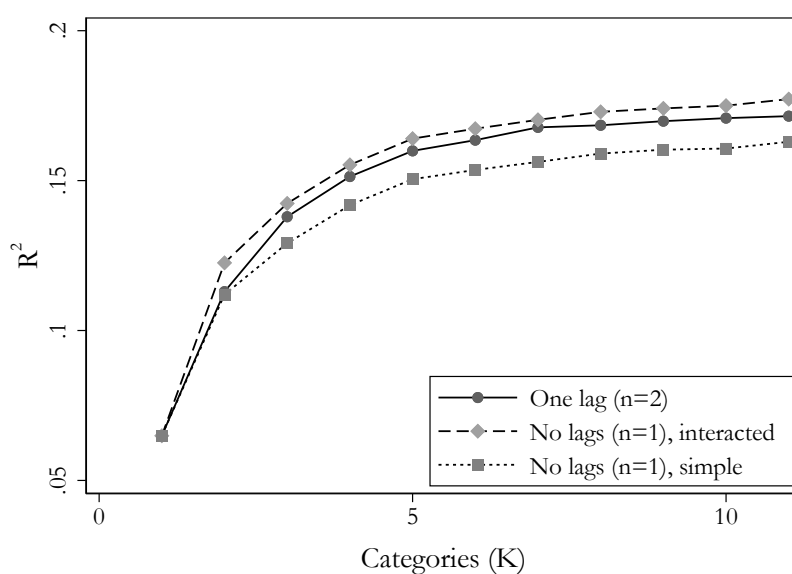


Figure A8: Performance of Alternative Risk Classifications: lags of classification.

Note: The figure displays *unadjusted* R^2 of a regression of expenditures on risk category indicators. All results are robust to using the *adjusted* R^2 . Each specification includes 21 age-gender fixed effects, year fixed effects and 79 plan fixed effects. Source: Own Calculations using German Claims Panel Data..

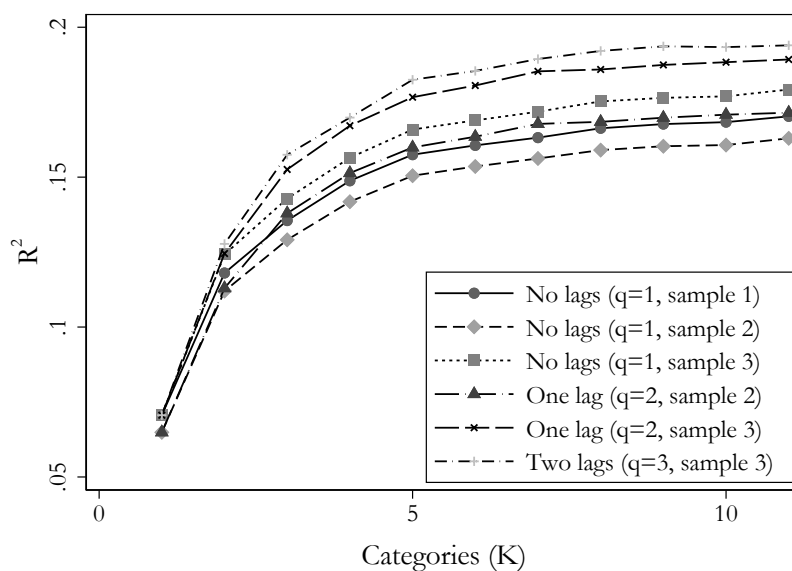


Figure A9: Performance of Alternative Risk Classifications: Different Samples.

Note: Each specification includes 21 age×gender fixed effects, year fixed effects and 79 plan fixed effects. Source: German Claims Panel Data.

Sample with low deductible. This robustness section focuses on plans with low deductibles. We consider a stricter sample selection rule, where we only include plans with deductibles below \$400.^{A70} These plans have approximately full coverage and thus more reliable information on the universe of health care expenditures. Summary statistics for this subsample are provided in Table A5. A comparison with the numbers in Table A1 makes clear that the two samples are very similar in terms of age, gender and history with the company. On the other hand, the restricted sample has a greater share of employees and civil servants, but a smaller share of self-employed. The plan characteristics are also similar to a great extent—with the obvious exceptions of deductible size and average claims.

Table A5: Summary Statistics: Low-Deductible Plans

	Mean	SD	Min	Max	N
Socio-Demographics					
Age (in years)	44.8	11.8	25.0	99.0	879,468
Female	0.256	0.437	0.0	1.0	879,468
Policyholder since (years)	7.7	5.3	1.0	40.0	879,468
Client since (years)	13.9	11.7	1.0	84.0	879,468
Employee	0.414	0.493	0.0	1.0	879,468
Self-Employed	0.281	0.449	0.0	1.0	879,468
Civil Servant	0.280	0.449	0.0	1.0	879,468
Health Risk Penalty	0.338	0.473	0.0	1.0	879,468
Pre-Existing Condition Exempt	0.015	0.121	0.0	1.0	879,468
Health Plan Parameters					
TOP Plan	0.342	0.475	0.0	1.0	879,468
PLUS Plan	0.397	0.489	0.0	1.0	879,468
ECO Plan	0.261	0.439	0.0	1.0	879,468
Annual premium (USD)	5,208	2,005	0	33,037	879,374
Annual risk penalty (USD)	133	347	0	21,214	879,468
Deductible(USD)	154	164	0	395	879,468
Total Claims (USD)	3,868	9,064	0	2,345,126	879,468

Source: German Claims Panel Data. *Policyholder since* is the number of years since the client has enrolled in the current plan; *Client since* is the number of years since the client joined the company. *Employee* and *Self-Employed* are dummies for the policyholders' current occupation. *Health Risk Penalty* is a dummy that is one if the initial underwriting led to a health-related risk add-on premium on top of the factors age, gender, and plan; *Pre-Existing Conditions Exempt* is a dummy which equals one if the initial underwriting led to a coverage exclusion of services for some conditions. The mutually exclusive dummies *TOP Plan*, *PLUS Plan* and *ECO Plan* capture the generosity of the plan. *Annual premium* is the annual premium, and *Annual Risk Penalty* is the amount of the health risk penalty charged. *Deductible* is the deductible and *Total Claims* the sum all claims in a calendar year. See Section 4.1 for further details.

Figure A10 compares the distributions of λ^* in the two samples. As expected, the zero-deductible

^{A70}This is the lowest cutoff for the deductible which gives us a sufficient number of observations to analyze health risk transitions within each age group.

plans have higher ACG[®] scores.

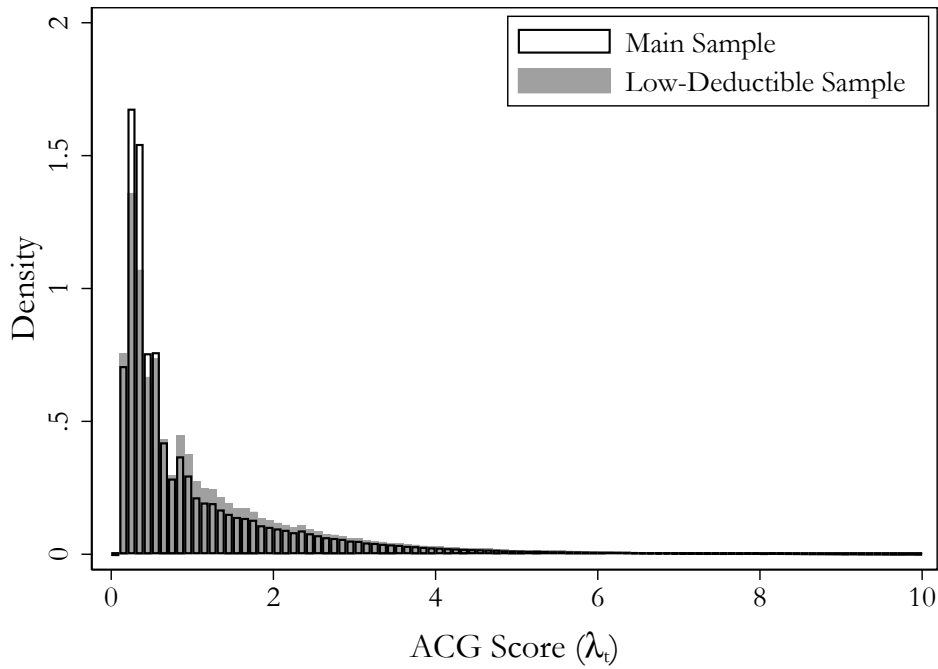


Figure A10: Distribution of λ^* for Main Sample vs. Low-Deductible Plans.

Table A6 shows how clients distribute over different risk categories by age in the low-deductible sample. A comparison with Table 2 confirms that the individuals in the low-deductible sample are in slightly worse health.

Table A6: Health Risk Categories λ by Age Group: Low-Deductible Sample

Age	1 (Healthiest)	2	3	4	5	6	7 (Sickest)
25-30	0.739	0.190	0.049	0.016	0.006	0.001	0.000
30-35	0.672	0.225	0.069	0.025	0.007	0.002	0.000
35-40	0.559	0.282	0.112	0.034	0.011	0.003	0.000
40-45	0.507	0.291	0.141	0.043	0.015	0.003	0.000
45-50	0.406	0.317	0.190	0.060	0.021	0.005	0.001
50-55	0.316	0.311	0.244	0.090	0.030	0.008	0.001
55-60	0.172	0.309	0.320	0.139	0.045	0.013	0.002
60-65	0.093	0.263	0.361	0.190	0.069	0.022	0.003
65-70	0.038	0.200	0.423	0.252	0.072	0.014	0.002
70-75	0.011	0.131	0.403	0.333	0.107	0.015	0.001
75+	0.000	0.055	0.286	0.453	0.179	0.024	0.003

Source: German Claims Panel Data. Sample includes all age groups and uses the ACG[®] score for the classification.

Table A7 shows the transition probabilities between different health states in the low-deductible sample. The probabilities are very similar to those reported in Table 1.

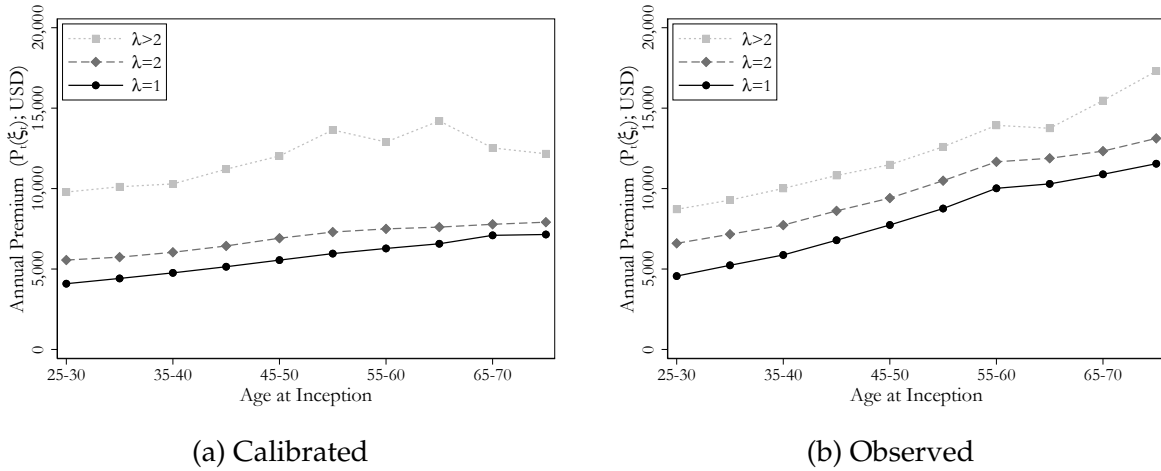
Table A7: Health Risk Category Transitions: Low-Deductible Sample

λ_t	λ_{t+1}							
	1	2	3	4	5	6	7	8 (+)
1	0.802	0.187	0.006	0.002	0.001	0.000	0.000	0.001
2	0.192	0.533	0.232	0.032	0.008	0.001	0.000	0.001
3	0.040	0.168	0.600	0.159	0.026	0.003	0.000	0.003
4	0.017	0.041	0.237	0.553	0.126	0.012	0.000	0.013
5	0.015	0.019	0.034	0.339	0.452	0.102	0.004	0.035
6	0.008	0.013	0.017	0.102	0.313	0.402	0.051	0.094
7	0.000	0.000	0.003	0.027	0.115	0.231	0.426	0.198

Source: German Claims Panel Data. Sample includes all years, all age groups, and uses the ACG[®] score for the classification.

A6 Observed vs. Calibrated GLTHI Premium Profiles

Figure A11 compares the (a) calibrated and (b) observed premium profiles for individuals entering their plan at different ages. In both figures, the highest category ($\lambda_t > 2$) is a weighted average calculated according to the actual distribution of λ_t in the different age groups.

**Figure A11: Calibrated vs. Actual Starting Premiums $P_t(\xi_t)$ by Age at Inception**

Source: German Claims Panel Data. In Figure A11 (b), the sample includes all years and all health plans, and clients who have been in their contract for 2 to 5 years. We adjusted premiums for the three benefit categories *TOP*, *PLUS*, *ECO* and deductible size.

A7 Quantifying Arrow Securities

As discussed in Section 3.3 dynamic long term contracts with one sided commitment can also be implemented by letting the individuals trade state-contingent one-period Arrow securities. Here we provide numerical examples for the GLTHI contract and for the optimal contract.

A7.1 GLTHI Contract

Consider an individual who enters the GLTHI market at age 25 in the healthiest state; $\xi_1 = \Xi_1 = 1$. As shown in Table 3, the GLTHI contract specifies a guaranteed-renewable premium of $P_1(1) = \$3,973$ which represents \$2,499 in excess of expected claims. This front-loading can be reinterpreted as the total amount paid for seven Arrow securities $b_1(k|\Xi_1 = 1)$, $k \in 1, \dots, 7$, purchased in period 1 and that would pay 1 dollar if the realized health state in period 2 is k . The prices of each security is given by the corresponding transition probability $\pi(k|1)$. Stacking quantities and probabilities in row vectors; i.e., $\mathbf{b}_1(1) \equiv (b_1(1|\Xi_1 = 1), \dots, b_1(7|\Xi_1 = 1))$, $\boldsymbol{\pi}_1(1) \equiv (\pi_1(1|\Xi_1 = 1), \dots, \pi_1(7|\Xi_1 = 1))$, we have:

$$\mathbf{b}_1(1) \times \boldsymbol{\pi}_1(1)' \times \delta + E(m_1|\xi_1 = 1) = P_1(1)$$

Applying formula 3 described in Lemma 1 to our data we get:

$$\mathbf{b}_1(1) = 10^3 \times (1.35, 11.88, 20.62, 28.90, 39.57, 60.25, 202.73)$$

In other words, to replicate the premium path under the GLTHI contract, an individual who starts in the healthiest state purchases a short-term contract in the spot market (at a premium $E(m_1|\xi_1 = 1) = 1,473$), plus a quantity $\mathbf{b}_1$ of seven Arrow securities, at a price $\boldsymbol{\pi}(1) \times \delta$.

Consider now an individual in period 2 with history $\Xi_2 \equiv (\xi_1, \xi_2) = (1, 1)$. As $P_1(1) < P_2(1)$, this individual does not lapse between period 1 and period 2, and continues to pay $\tilde{P}_2([1, 1]) = \$3,973$ in period 2. The individual buys securities in the amount of $\mathbf{b}_2(\xi_2 = 1|\Xi_1 = 1)$ such that

$$\mathbf{b}_2(\xi_2 = 1|\Xi_1 = 1) \times \boldsymbol{\pi}_2(1)' \times \delta + E(m_2|\xi_2 = 1) - b_1(1|\Xi_1 = 1) = \tilde{P}_2([1, 1]),$$

where $\mathbf{b}_2(\xi_2 = 1|\Xi_1 = 1)$ and $\boldsymbol{\pi}_2(1)$ are analogously defined as $\mathbf{b}_1(1)$ and $\boldsymbol{\pi}_1(1)$, respectively. We get $\mathbf{b}_2(\xi_2 = 1|\Xi_1 = 1) = 10^3 \times (2.74, 13.25, 22.04, 30.41, 41.22, 62.19, 192.92)$.

Note that the amount of securities bought in any given period t for every future realization of the health status depends on age and on the full history of health statuses up to t .

A7.2 Optimal Contracts

Using the results in A3.1, the vector $b_1^{GHHW}(1)$ shows the quantities of securities purchased for the period-2 contingencies given the individual starts in period 1 that replicate the optimal contract.

$$b_1^{GHHW}(1) = 1000 \times \begin{bmatrix} 0.00 & 3.36 & 8.88 & 15.68 & 25.55 & 44.70 & 181.93 \end{bmatrix}$$

The vector $b_2^{GHHW}([1, 1])$ shows the quantities of securities that an individual who starts in period 1 in health state $\xi_1 = 1$, and transitions to state $\xi_2 = 1$ buys for the contingency of transitioning to state $\xi_3 = k$ in period 3.

$$b_2^{GHHW}([1, 1]) = 1000 \times \begin{bmatrix} 0.00 & 3.36 & 8.92 & 15.79 & 25.76 & 45.13 & 172.15 \end{bmatrix}$$

A8 Lapsation Rates

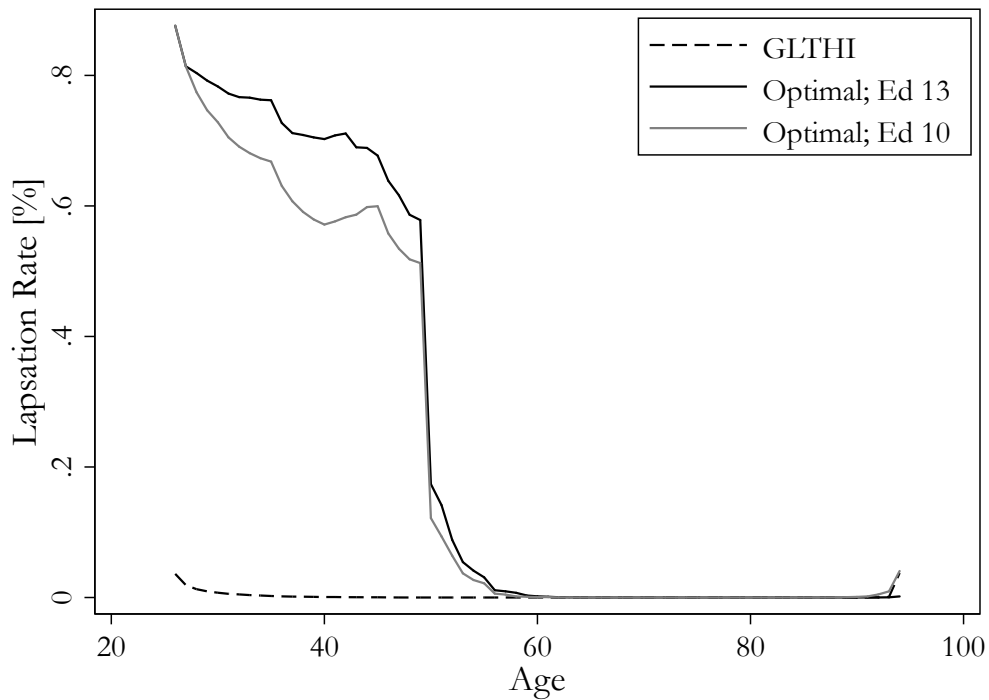


Figure A12: Lapsation Rates over the Lifecycle for Different Contracts

Figure A12 compares average lapsation rates under each long-term contract.^{A71} As expected, lapsation from GLTHI is extremely low over the entire lifecycle. In contrast, when expected future health improves, the optimal contract results in higher consumption for the healthiest types (and therefore for sicker types too) early in life. Lapsation in the optimal contract decreases substantially in the late 40s. At this point, most individuals have achieved their consumption plateau. Subsequently, consumption remains constant in order to transfer resources intertemporally and to save for old age.

^{A71}Lapsing under the optimal contract is defined as an increase in the consumption guarantees. As noted by Ghili et al. (2024), optimal contracts impose a “no-lapsation constraint”, so that the consumer will always stay in the same contract. However, an increase in the consumption guarantee specified within a contract can also be interpreted as a lapsation from an equivalent set of guaranteed premium paths. Figure A12 uses this interpretation of lapsing.

A9 Welfare Concepts

We use the concept of lifetime utility U to quantify welfare following, e.g., [Ghili et al. \(2024\)](#):

$$U = \mathbb{E} \left(\sum_{t=t_0}^T S_t \delta^{t-t_0} u(c_t) \right)$$

where S_t is an indicator of survival until period t , and c_t is the consumption in period t that is specified by the contract. It may depend on the history of health and income realizations up to t . Expectation is taken over the individual's lifetime health history $(\xi_1, \xi_2, \dots, \xi_t)$ and survival.

Certainty Income Equivalent. With a parametric assumption for flow utility $u(\cdot)$, and knowing income y_t , we can summarize welfare with the “certainty income equivalent”, denoted CE , such that:

$$u(CE) = \frac{\mathbb{E} \left(\sum_{t=t_0}^T S_t \delta^{t-t_0} u(c_t) \right)}{\mathbb{E} \left(\sum_{t=t_0}^T S_t \delta^{t-t_0} \right)} \quad (12)$$

This simple expression captures the main trade-offs in health insurance design for lifetime welfare. Lifetime utility is higher when consumption is smoothed across health states and across periods.

First-Best. In particular, the *first-best* consumption level equals the annualized present discounted value of “net income” $y_t - \mathbb{E}(m_t)$, taking into account mortality risk.^{A72} This constant optimal consumption level C^* is given by:

$$C^* = \frac{\mathbb{E} \left(\sum_{t=t_0}^T S_t \delta^{t-t_0} (y_t - \mathbb{E}(m_t)) \right)}{\mathbb{E} \left(\sum_{t=t_0}^T S_t \delta^{t-t_0} \right)} \quad (13)$$

Short-Term Contracts. Under a series of actuarially fair *short-term* contracts, the premium in period t with health status ξ_t will simply be $\mathbb{E}(m_t)$. Thus consumption will be $c_t = y_t - \mathbb{E}(m_t|\xi_t)$, and the certainty equivalent CE becomes:

$$u(CE_{ST}) = \frac{\mathbb{E} \left(\sum_{t=t_0}^T S_t \delta^{t-t_0} u(y_t - \mathbb{E}(m_t|\xi_t)) \right)}{\mathbb{E} \left(\sum_{t=t_0}^T S_t \delta^{t-t_0} \right)} \quad (14)$$

^{A72}We assume that there is no annuity market, so mortality risk is still considered.

A10 Additional Material for Robustness Analysis

A10.1 Epstein-Zin Recursive Preferences

In the robustness section we evaluate welfare under Epstein-in preferences. Here we provide the derivation for the formula we use to compute the certainty consumption equivalent with Epstein-Zin preferences.

Preferences are defined recursively as

$$V_t = F(c_t, R_t(V_{t+1})),$$

with $R_t(V_{t+1}) = G^{-1}(\mathbb{E}_t G(V_{t+1}))$. As mentioned in the main text, we use the CES aggregator for $F(c, z) = ((1 - \delta)c^{1-1/\psi} + \delta z^{1-1/\psi})^{\frac{1}{1-1/\psi}}$, and incorporate the CARA utility function as $G(c) = u(c) = \frac{1}{\gamma} e^{-\gamma c}$.

Throughout we have assumed that utility is zero if the individual is dead. We can re-interpret V_t as the value of being alive in period t . Under that interpretation, one can write preferences recursively as:

$$V_t = \left((1 - \delta) c_t^{1-1/\psi} + s_t \delta R_t(V_{t+1})^{1-1/\psi} \right)^{\frac{1}{1-1/\psi}} \quad (15)$$

where s_t is the probability of survival between t and $t + 1$.

We now derive an expression for the certainty equivalent consumption c for any given value V_t under recursive preferences. Consider the situation in which consumption (while alive) is constant and equal to c . This means that $R_t(V_{t+1}) = V_{t+1}$, and therefore we can re-write

$$V_t = \left((1 - \delta) c^{1-1/\psi} + s_t \delta (V_{t+1})^{1-1/\psi} \right)^{\frac{1}{1-1/\psi}} \quad (16)$$

Replacing the V_{t+1} in Equation (16) as a function of V_{t+2} yields

$$\begin{aligned} V_t &= \left((1 - \delta) c^{1-1/\psi} + s_t \delta \left((1 - \delta) c^{1-1/\psi} + \delta s_{t+1} (V_{t+2})^{1-1/\psi} \right) \right)^{\frac{1}{1-1/\psi}} \\ &= \left((1 - \delta) c^{1-1/\psi} + s_t \delta (1 - \delta) c^{1-1/\psi} + s_t s_{t+1} \delta^2 V_{t+1}^{1-1/\psi} \right)^{\frac{1}{1-1/\psi}} \end{aligned}$$

Iterating forward we can show that

$$\frac{V_t^{1-1/\psi}}{1 - \delta} = \sum_{j=t}^T c^{1-1/\psi} \delta^{j-t} S_t^j$$

where $S_t^j \equiv \prod_{k=t}^j s_k$ is the survival probability from t to j . Solving for c , we get an expression defining the certainty equivalent:

$$c = \left(\frac{V_t^{1-1/\psi}}{1-\delta} \right)^{\frac{1}{1-1/\psi}} \left(\sum_{j=t}^T \delta^{j-t} S_t^j \right)^{-1} \quad (17)$$

Equation (17) provides the certainty equivalent consumption to a program that provides value V_t .

We are interested in the certainty equivalent taking into account the uncertainty regarding the “birth state” ξ_{t_0} . Denote the value of this lottery V_b . It can be expressed as a function of V_{t_0} (the value at age 25):

$$V_b = G^{-1}(\mathbb{E}_0(G(V_{t_0}(\xi_{t_0})))) \quad (18)$$

where $\mathbb{E}_0(\cdot)$ takes expectations with respect to the uncertain “birth” state, ξ_{t_0} .

For each contract, we can compute the value $V_{t_0}(\xi_{t_0})$, for each state ξ_{t_0} , *via* backwards induction. Plugging Equation (18) into Equation (17), applied to the initial period t_0 we get that the consumption certainty equivalent can be expressed as:

$$c = \left(\frac{\left(G^{-1}(\mathbb{E}_0(G(V_{t_0}(\xi_{t_0})))) \right)^{1-1/\psi}}{1-\delta} \right)^{\frac{1}{1-1/\psi}} \left(\sum_{j=t_0}^T \delta^{j-t_0} S_{t_0}^j \right)^{-1}, \quad (19)$$

where $\mathbb{E}_0(\cdot)$ takes expectations with respect to the “birth” state, ξ_{t_0} and S_t^j is the survival probability from t to j . For each contract, we compute $V_{t_0}(\xi_{t_0})$ numerically *via* backwards induction.

A10.2 Initial Health Status Representative of Germany.

In the robustness section we evaluate welfare using a health risk profile that is more representative of the German population as a whole. As mentioned in Appendix A, there is selection into GLTHI contracts for a number of reasons, some imposed by law and some resulting from private incentives of clients and insurance providers. For this reason, it is of great interest to understand if and how our welfare results change when we consider the GKV population.

We start by characterizing and contrasting the two populations, where we draw on two distinct datasets: the representative SOEP panel and a GKV claims dataset. The latter includes a random 5% sample from 2010-11 from one of the largest GKV insurers, 5.6 million enrollees.

Table A8: Socio-Demographics of GKV versus PKV Population

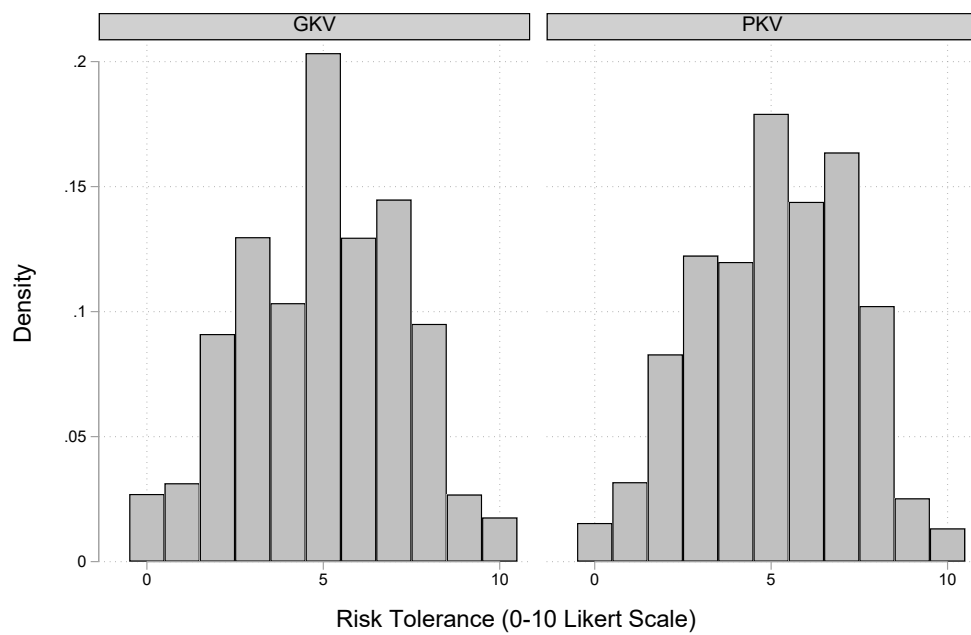
	GKV	PKV	PKV			
			civil servants	white collar	self- employed	non- working
Age	30.110	30.802	30.464	31.540	31.608	29.615
Female	0.467	0.414	0.570	0.195	0.216	0.600
Full-time	0.625	0.817	0.908	0.891	0.931	0.000
Part-time	0.166	0.087	0.092	0.109	0.069	0.000
Dropout	0.017	0.011	0.019	0.000	0.010	0.000
High school	0.315	0.643	0.705	0.723	0.410	0.709
Monthly gross wage	2,611	3,773	3,104	5,285	3,884	2,820
Monthly net wage	1,671	2,581	2,396	3,196	2,466	2,595
Annual household income	24,575	34,858	30,620	46,927	35,540	28,729
Risk tolerance (0-10)	5.040	5.355	4.808	5.885	6.301	4.620
Smoker	0.395	0.292	0.300	0.222	0.385	0.162
BMI	25.032	23.995	23.581	24.536	24.453	23.350
Physical issues, accomplishes less	0.064	0.036	0.019	0.016	0.090	
Physical limitations	0.053	0.031	0.021	0.012	0.025	0.117
Emotional issues, accomplishes less	0.053	0.030	0.022	0.017	0.012	0.153
Emotional limitations	0.033	0.015	0.012	0.001	0.004	0.085
Hospital stay last year	0.097	0.078	0.073	0.050	0.050	0.235
Hospital nights last year	0.804	0.445	0.374	0.302	0.178	1.613
Doctor visits last quarter	1.780	1.775	2.036	1.247	1.399	2.588
N	23,250	2282	997	465	503	262

Notes: The sample includes 25 to 35 year olds from 2004 to 2016 (except 2005 and 2007). SOEP-provided weights are applied. The number of observations is small for the health behavior, health, and health care utilization measures as they were not surveyed in all years. All income measures have been consistently generated and cleaned by the SOEP team; e.g., *Monthly gross wage* is labeled *labgro* and *Monthly net wage* is labeled *labnet* in [SOEP \(2018\)](#). All income measures are in \$2016. When applying t-tests, all means for GKV and PKV are statistically different from one another at conventional levels. Source: [SOEP \(2018\)](#). See Section 4.2 for a detailed discussion of the dataset.

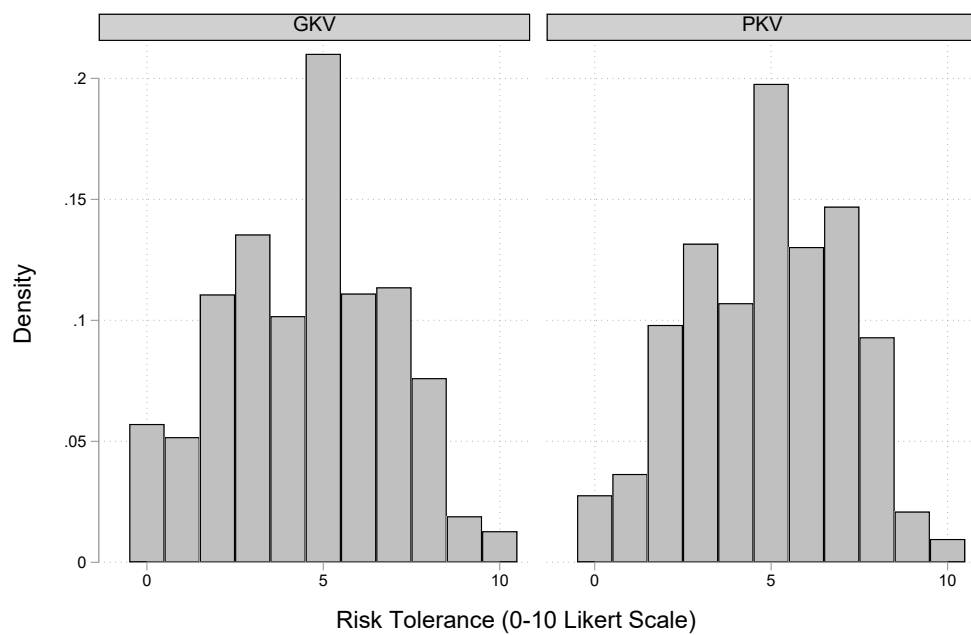
Table A8 provides descriptive statistics from the SOEP, focusing on the 25-35 year-olds. It confirms that the PKV population is positively selected in terms of health and various socioeconomic outcomes. Figure A13 plots the distribution of risk tolerance in the two groups. Even if the risk tolerance is higher among the privately insured on average (5.4 versus 5.0) there is a great deal of overlap of the two distributions, for younger individual as well as for the entire population.

Table A9 compares summary statistics from our PKV claims data to GKV claims data. The GKV sample is seven years older than the PKV sample. The ACG score is also larger on average. When we consider the subsample of 25-35-year-olds in the two rightmost columns, the difference in ACG scores shrinks, but remains larger for the GKV sample. Figure A14 shows the distribution of 2011

ACG scores in the GKV and PKV samples; the left panel plots the raw data for both; the right panel re-weights the GKV observations in order to achieve the same distribution of age by gender as in the PKV dataset.



(a) 25-35 Year Olds



(b) All adults

Figure A13: Risk Tolerance: GKV versus PKV population

Notes: The figure shows the distribution of Risk-Tolerance (in a 0-10 Likert Scale) for GKV (public) and PKV (private) enrollees. Panel (a) shows the results for young adults (25-35 year olds) and Panel (b) shows the results for all adults. Source: [SOEP \(2018\)](#).

Table A9: Summary Statistics: PKV versus GKV Claims Data, 2011

	ALL AGED 25+		AGE 25 – 35	
	GKV	PKV	GKV	PKV
Age (in years)	56.3	48.9	29.9	32.2
Female	0.648	0.290	0.558	0.246
ACG Score	2.886	1.745	1.233	0.928
N	226,054	96,511	28,795	9,857

Notes: GKV statistics are based on a 5% random sample of a population of 5.6 million enrollees. Source: German Claims Panel Data and GKV data.

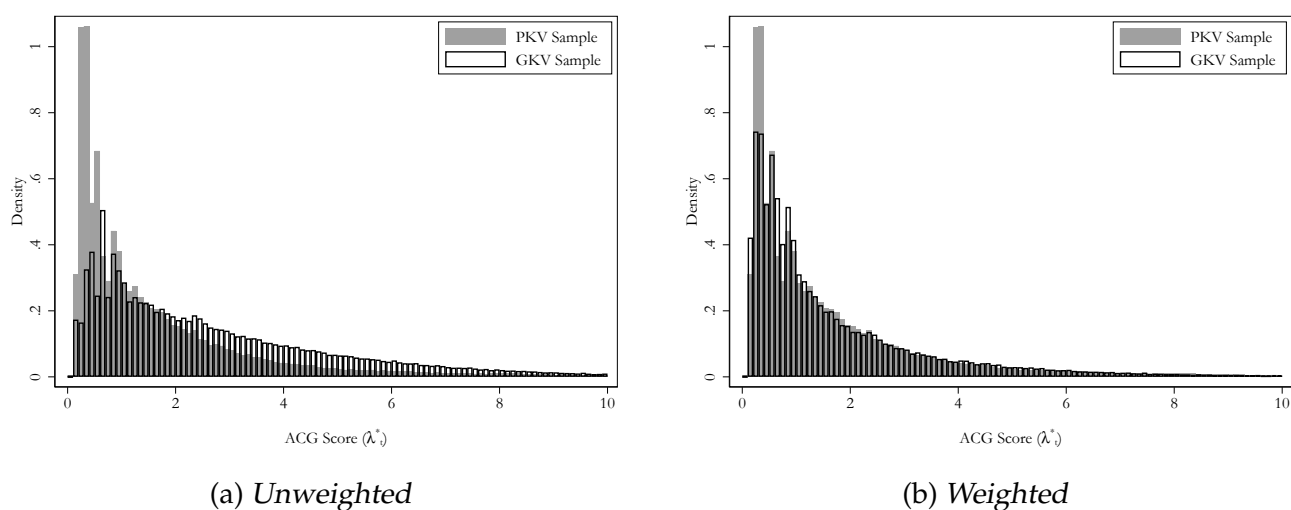


Figure A14: Distribution of 2011 ACG Scores: GKV versus PKV population

Note: The figures show histograms of 2011 ACG scores for the GKV and the PKV sample. Panel (a) plots the raw data and panel (b) re-weights the GKV data to get identical distributions of age and gender. Both samples are truncated at an ACG score of 10.

Table A10: Health Risk Categories λ by Age Group: GKV Sample, 2011

Age	1 (Healthiest)	2	3	4	5	6	7 (Sickest)
25-30	0.474	0.316	0.143	0.052	0.012	0.002	0.001
30-35	0.423	0.325	0.161	0.069	0.018	0.004	0.001
35-40	0.296	0.364	0.216	0.088	0.028	0.007	0.001
40-45	0.262	0.366	0.230	0.099	0.033	0.009	0.002
45-50	0.153	0.366	0.290	0.130	0.046	0.013	0.002
50-55	0.127	0.311	0.309	0.165	0.067	0.018	0.003
55-60	0.038	0.254	0.350	0.228	0.097	0.029	0.005
60-65	0.024	0.204	0.333	0.265	0.127	0.042	0.006
65-70	0.025	0.115	0.357	0.330	0.147	0.022	0.003
70-75	0.005	0.069	0.306	0.377	0.205	0.035	0.003
75+	0.000	0.025	0.172	0.407	0.322	0.068	0.005

Notes: The table shows the fraction of individuals in each category by age group. The sample includes all years, all age groups, and uses the ACG scores to construct risk categories λ as explained in the text. Source: Own calculations based on GKV Data.

Table A11 compares our baseline welfare results (Panel (h)) to those of a GKV-only population based on their distribution over starting states (Panel (i)) and to a population approximating the real-world mix of 90% GKV and 10% PKV enrollees (Panel(j)).

Table A11: Welfare under different contracts with initial states representatives of Germany

	(1) C^*	(2) C_{ST}	(3) C_{GHHW}	(4) C_{GLTHI}	(5) $\frac{C_{GHHW}-C_{GLTHI}}{C_{GHHW}}$
Panel (i): $\Delta_0 = \frac{1}{100} [47.47, 31.58, 14.33, 5.16, 1.21, 0.24, 0]$					
Ed 10	22,700	-10,410	19,228	18,966	1.4%
Ed 13	33,878	-2,839	21,633	21,446	0.9%
Panel (j): $\Delta_0 = \frac{1}{100} [51.63, 29.45, 12.95, 4.65, 1.10, 0.22, 0]$					
Ed 10	22,728	-10,476	19,443	19,155	1.5%
Ed 13	33,906	-2,969	21,885	21,677	0.9%

Notes: The table shows welfare measured by the consumption certainty equivalents in 2016 USD dollars, per capita, per year, separately for two income profiles (see Figure 3). Panel (i) uses the estimated initial probabilities at 25 for the GKV population, and Panel (j) uses a mixture including 90% of individuals in GKV and 10% of individuals in PKV. Columns (1) to (4) show welfare according to the (1) first-best (C^*), (2) a series of short-term contracts (C_{ST}), (3) the optimal contract (C_{GHHW}), and (4) the GLTHI (C_{GLTHI}). Column (5) and (6) show how much of the welfare gap between (2) and (1) is closed by the optimal contract and the GLTHI, respectively. Column (7) shows the fraction of the potential benefits from long-term contracting achieved by GLTHI. Column (8) shows the percentage of welfare loss under GLTHI relative to the optimal contract. Source: Own calculations based on German Claims Panel Data, SOEP data.

A10.3 Different Distribution of Starting States

In this section, we assess the robustness of our main results (Table 4) with respect to the initial distribution of starting states. Specifically, we sampled 20 million probability simplices $\tilde{\Delta}_r \in \Delta^7$ from a Dirichlet distribution with concentration parameters equal to the baseline probabilities coming out of the risk classification procedure (including the worst health state 7). The resulting probability simplices span a wide range of expected costs.

Figure A15 plots the resulting welfare difference between GLTHI and the optimal contract in relation to the expected costs associated with each draw. The point “Baseline” corresponds to our baseline estimate in Table 4. According to Figure A15, the welfare loss is bounded from above at about 6% (4% for the less-educated group). The maximum welfare loss decreases in expected expenditures.

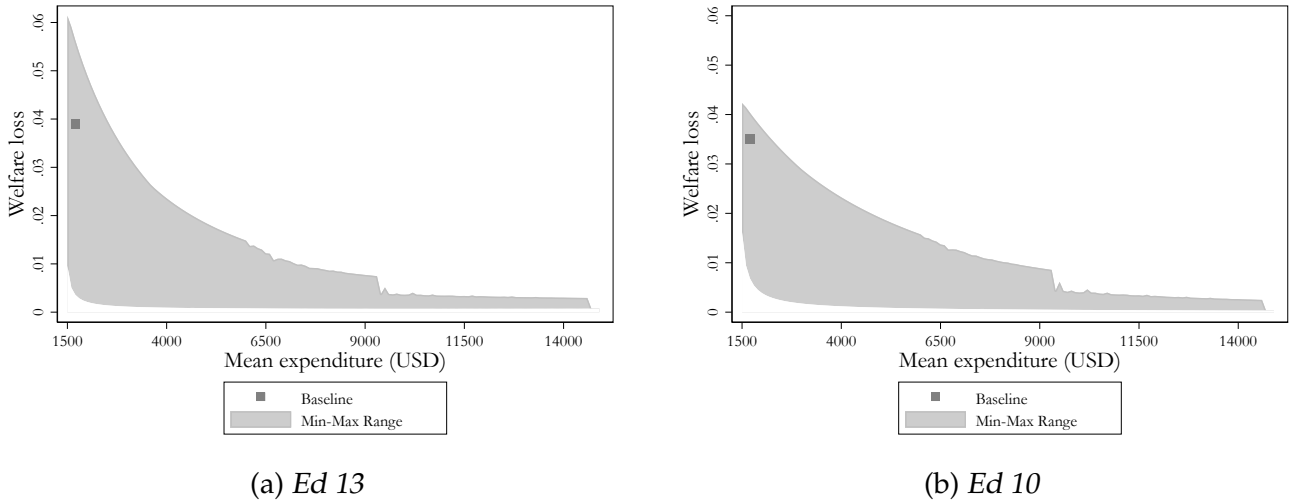


Figure A15: Sensitivity Analysis: Distribution of Starting States.

Note: The figures show maxima and minima of GLTHI welfare losses ($\frac{C_{GHHW} - C_{GLTHI}}{C_{GHHW}}$) within increments of \$100 of expected expenditure. The underlying distribution is based on 20 million draws from a Dirichlet distribution.

A10.4 Allowing for Savings

In the robustness section we allow for savings under short-term contracts and under the GLTHI contract. Here we provide details of the optimal savings problem.

Individuals solve the following maximization problem:

$$\begin{aligned}
& \max_{c_t} \quad \mathbb{E} \left(\sum_{t=t_0}^T S_t \delta^t u(c_t) \right) \\
& \text{s.t.} \quad a_{t_0} = 0 \\
& \quad \quad a_t \geq 0 \quad \forall t \\
& \quad \quad a_{t+1} = (1+r)a_t + y_t - c_t - \tilde{P}_t(\Xi_t)
\end{aligned}$$

where $\tilde{P}_t(\Xi_t)$ is the premium paid in period t as a function of an individual's medical history $\Xi_t \equiv (\xi_1, \xi_2, \dots, \xi_t)$, and a_t is the level of assets.

Different contracts result in different mappings between an individual's medical history up to period t and an individual's premium in t . Under a series of short-term contracts, only an individual's current health status matters since $\tilde{P}_t(\Xi_t) = \mathbb{E}(m_t | \Xi_t) = \mathbb{E}(m_t | \xi_t)$. In contrast, for a GLTHI contract, the entire medical history matters. Due to guaranteed-renewability, $\tilde{P}_t(\Xi_t)$ is defined recursively: In the first period, $\Xi_1 = \xi_1$ and $\tilde{P}_1(\Xi_1) = P_1(\xi_1)$, where Equation (1) in the text defines $P_t(\xi_t)$. In any period $t > 1$, $\tilde{P}_t(\Xi_t) = \min\{\tilde{P}_t(\Xi_{t-1}), P_t(\xi_t)\}$. The state variable in the dynamic program under GLTHI is the guaranteed-renewable premium; its law of motion is given by the probability of qualifying for a lower premium. Note that, in this optimal consumption problem with savings, there is uncertainty regarding net income $y_t - \tilde{P}_t(\Xi_t)$ and mortality risk. Mortality risk implies that individuals may die with positive assets. Therefore, the expected net present value of consumption with optimal savings will be lower than the net present value of resources. Our calculations implicitly assume that individuals do not derive value from bequests.

For a given lifecycle income profile, the dynamic program provides an optimal consumption policy $C_t^*(\xi_t, a_t)$ where a_t is the level of assets carried into period t . The certainty equivalent (CE) of the dynamic problem is equal to:

$$u(C_{SAV}) = \frac{\mathbb{E} \left(\sum_{t=t_0}^T S_t \delta^{t-t_0} u(C_t^*(\xi_t, a_t)) \right)}{\mathbb{E} \left(\sum_{t=t_0}^T S_t \delta^{t-t_0} \right)} \quad (20)$$

A10.5 US life-cycle income profiles

To analyze the robustness of our results with respect to using US life-cycle income profile, we employ the representative Panel Study of Income Dynamics (PSID). Figure A16 shows the results of estimating Equation 6 using PSID data for high-school and college-educated. The profiles shown in the Figure are inputs for the welfare calculation using US life-cycle income profiles as described in the main text.

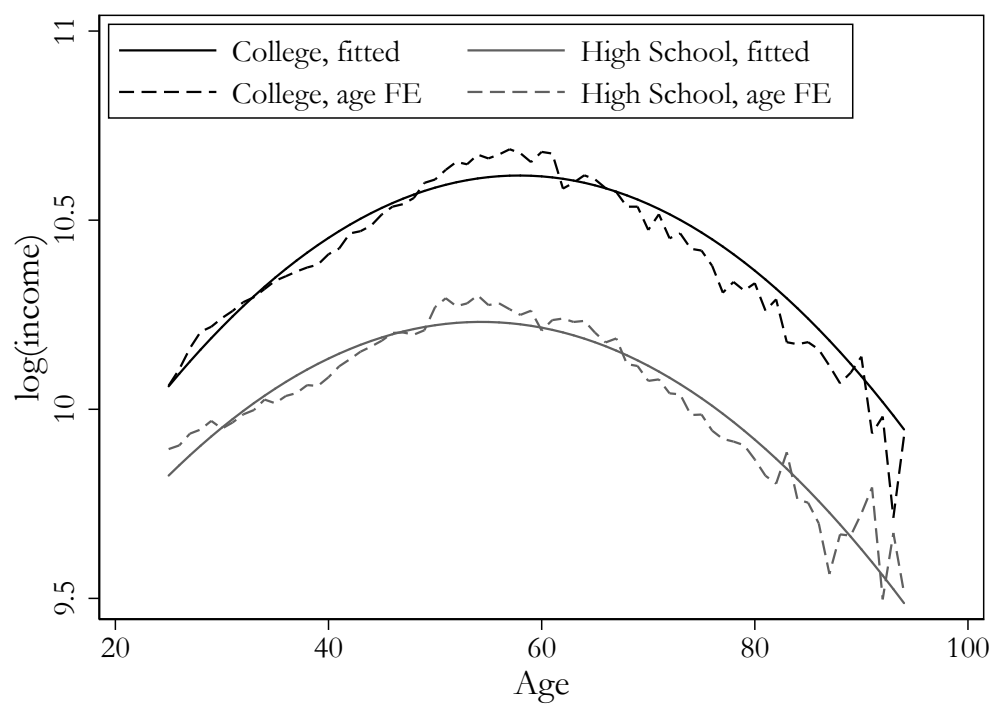


Figure A16: Lifecycle Income Paths for the United States, Nonparametric and Fitted.

Source: [Panel Study of Income Dynamics \(2018\)](#); [Frick et al. \(2007\)](#), years 1984 to 2015. All values in 2016 USD.